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(54) **NUMEROLOGY FOR 60GHZ OPERATION
TO ACHIEVE CHANNELIZATION
ALIGNMENT**

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H04L 27/26 (2006.01)

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CPC **H04W 72/0453** (2013.01); **H04L 27/2601**
(2013.01)

(58) **Field of Classification Search**
CPC H04L 27/2601; H04W 72/0453
See application file for complete search history.

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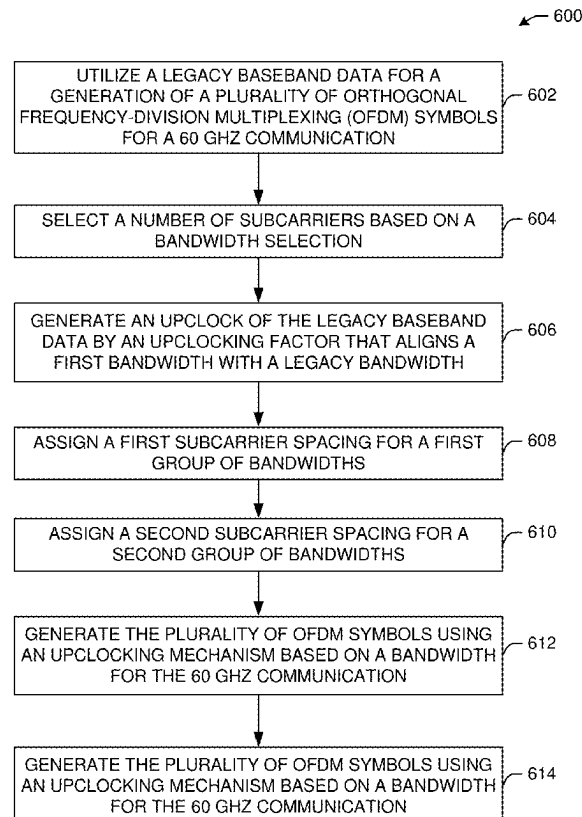
Primary Examiner — James P Duffy

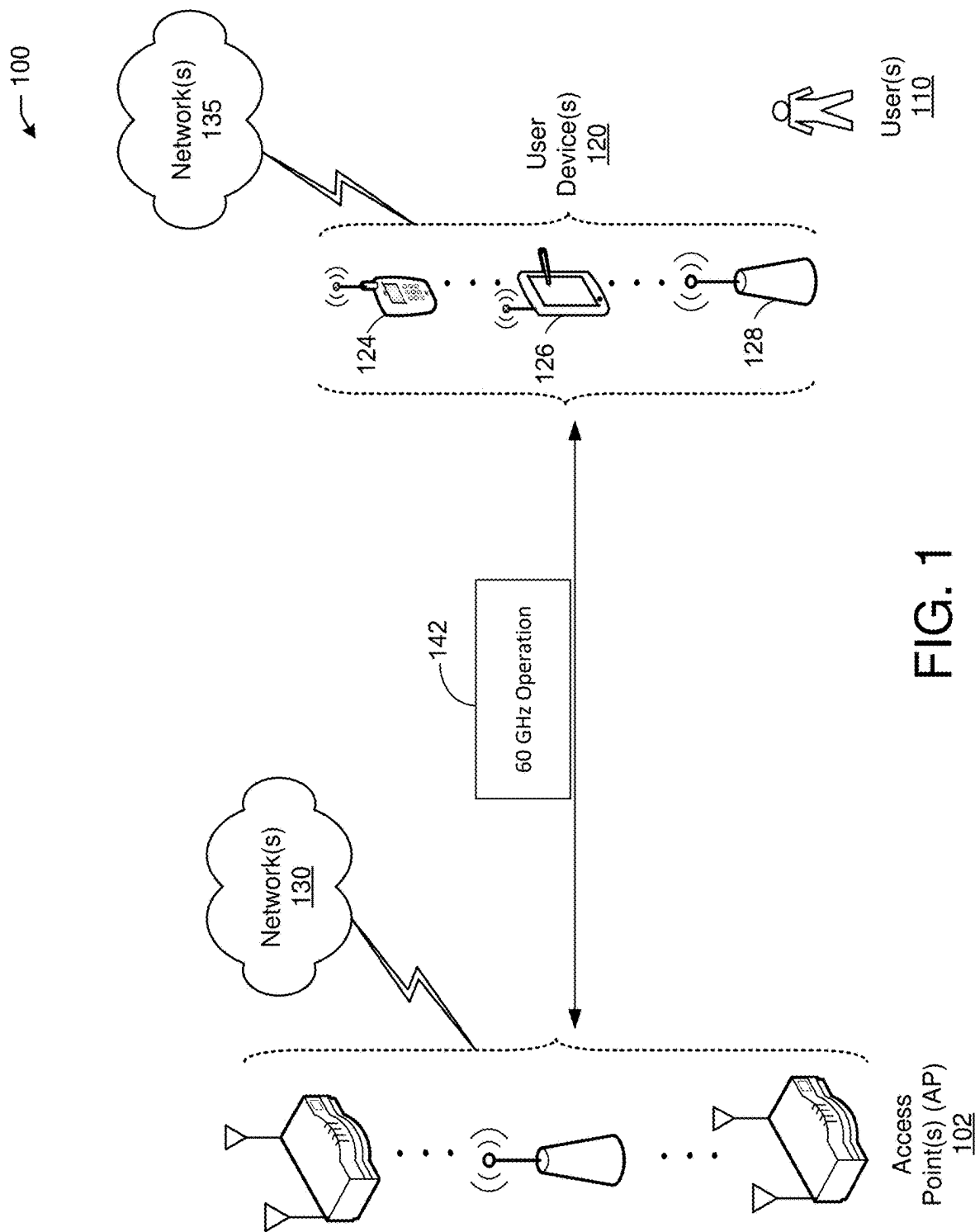
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(57) **ABSTRACT**

This disclosure describes systems, methods, and devices related to 60 GHz operation. A device may utilize a legacy baseband data for a generation of a plurality of orthogonal frequency-division multiplexing (OFDM) symbols for a 60 GHz communication. The device may select a number of subcarriers based on a bandwidth selection. The device may generate an upclock of the legacy baseband data by an upclocking factor that aligns a first bandwidth with a legacy bandwidth. The device may assign a first subcarrier spacing for a first group of bandwidths. The device may assign a second subcarrier spacing for a second group of bandwidths. The device may generate the plurality of OFDM symbols using an upclocking mechanism based on a bandwidth for the 60 GHz communication. The device may cause to send a plurality of OFDM symbols to a station device.

20 Claims, 12 Drawing Sheets





Legacy (11ac)

	20	40	80	160
N_{DFT}	64	128	256	512
ΔF	312.5KHz			
T_{DFT}	3.2 μ s			
T_s	50ns	25ns	12.5ns	6.25ns
T_{GI}	$TDFT/4 = 0.8\mu$ s			
N_{GI}	16	32	64	128
$T_{GI\ short}$	$TDFT/8 = 0.4\mu$ s			
$N_{GI\ short}$	8	16	32	64

6.75 x

	135	270	540	1080
N_{DFT}	64	128	256	512
ΔF	2.11MHz			
T_{DFT}	0.4 μ s			
T_s	6.25ns	3.12ns	1.56ns	0.78ns
T_{GI}	$TDFT/4 = 100$ ns			
N_{GI}	16	32	64	128
$T_{GI\ short}$	$TDFT/8 = 50$ ns			
$N_{GI\ short}$	8	16	32	64

FIG. 2

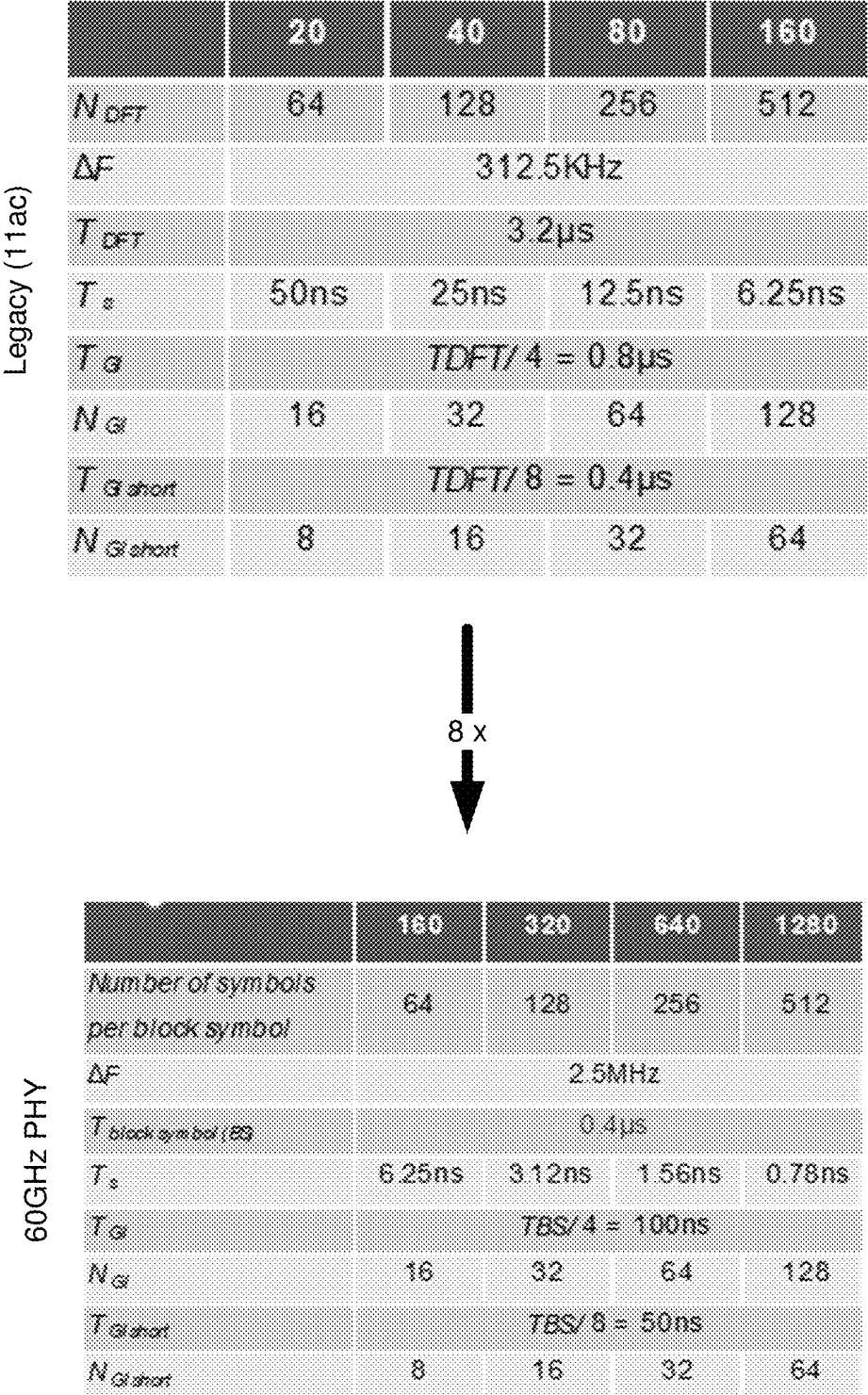


FIG. 3

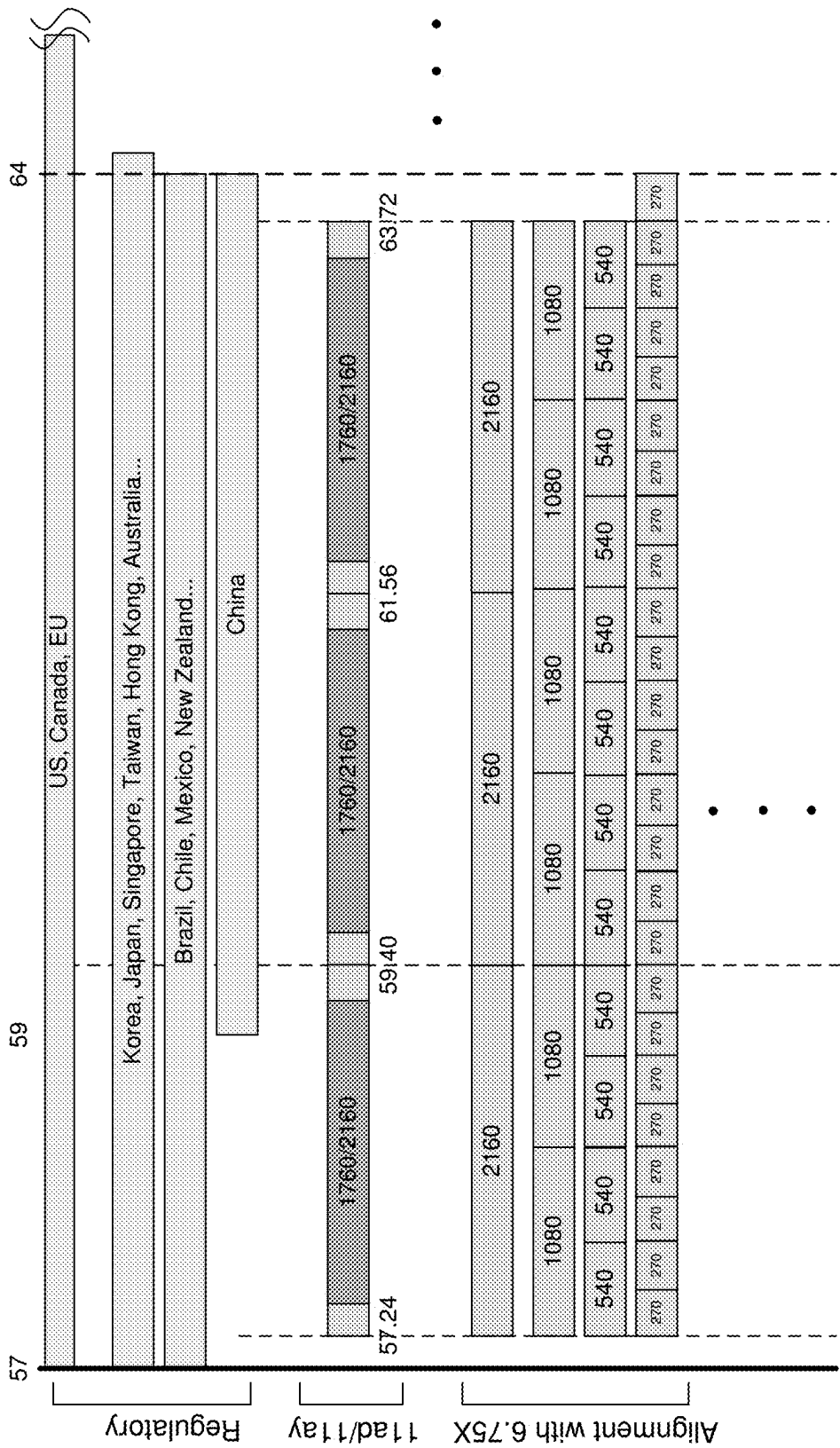
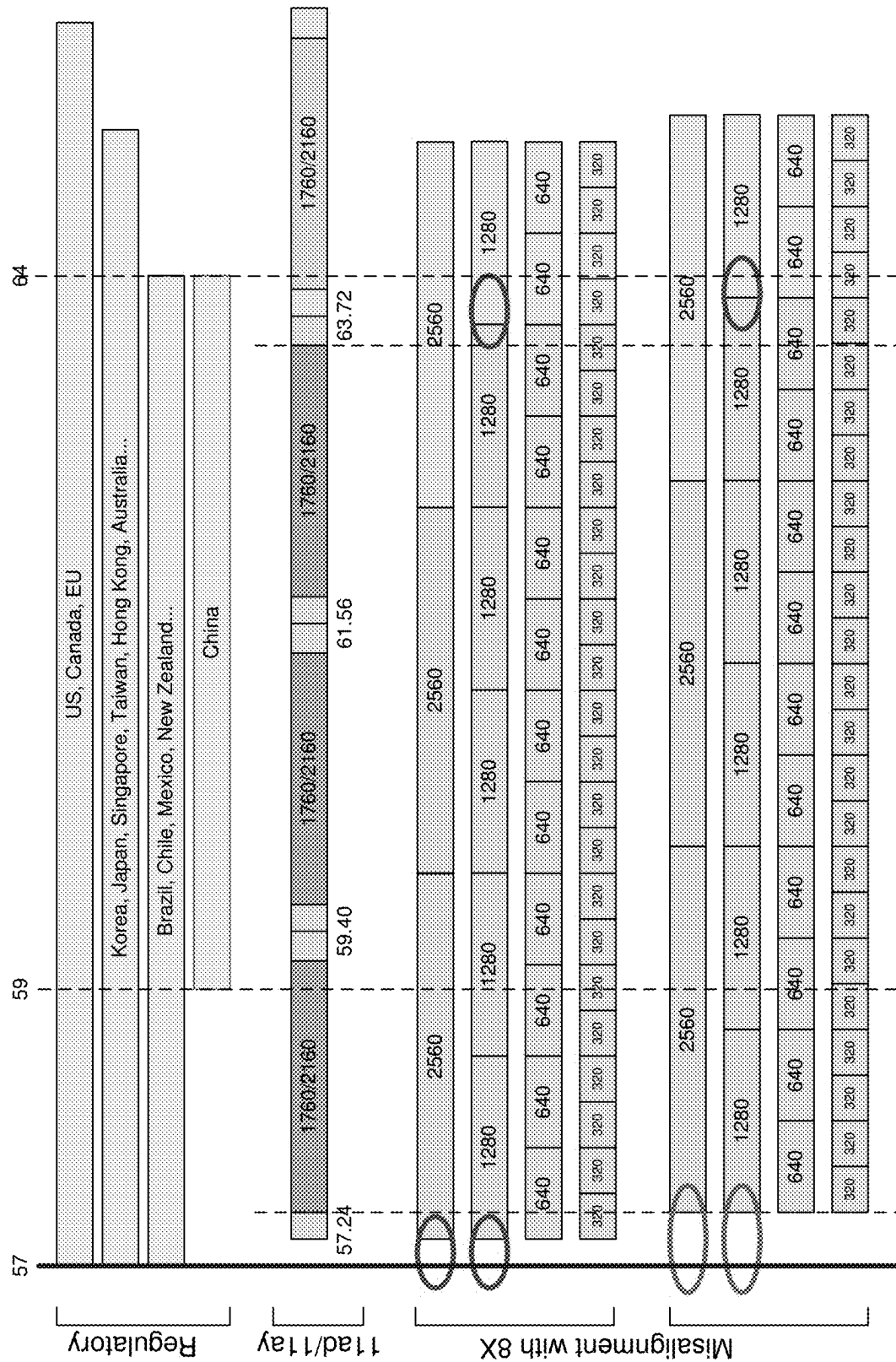


FIG. 4



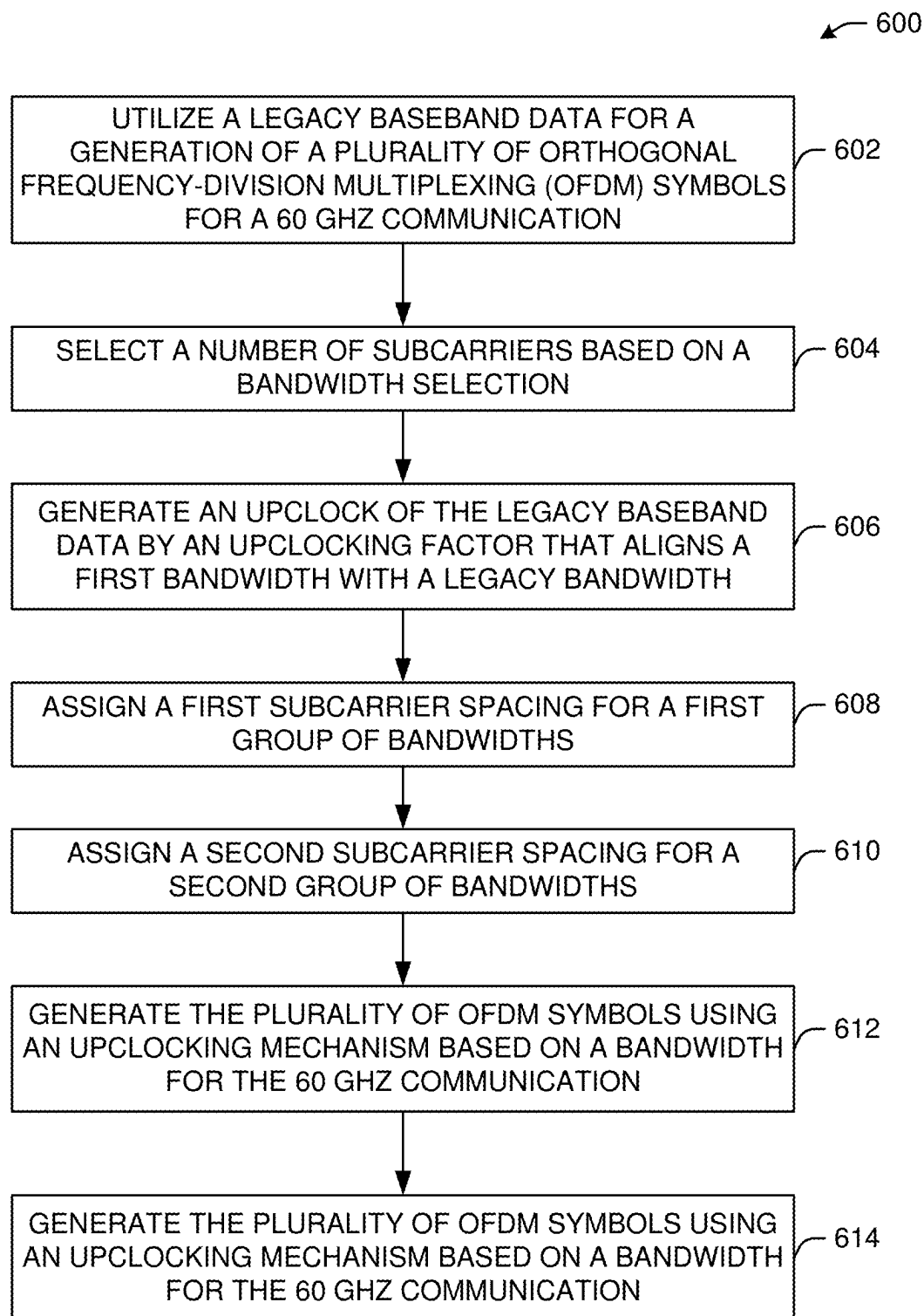


FIG. 6

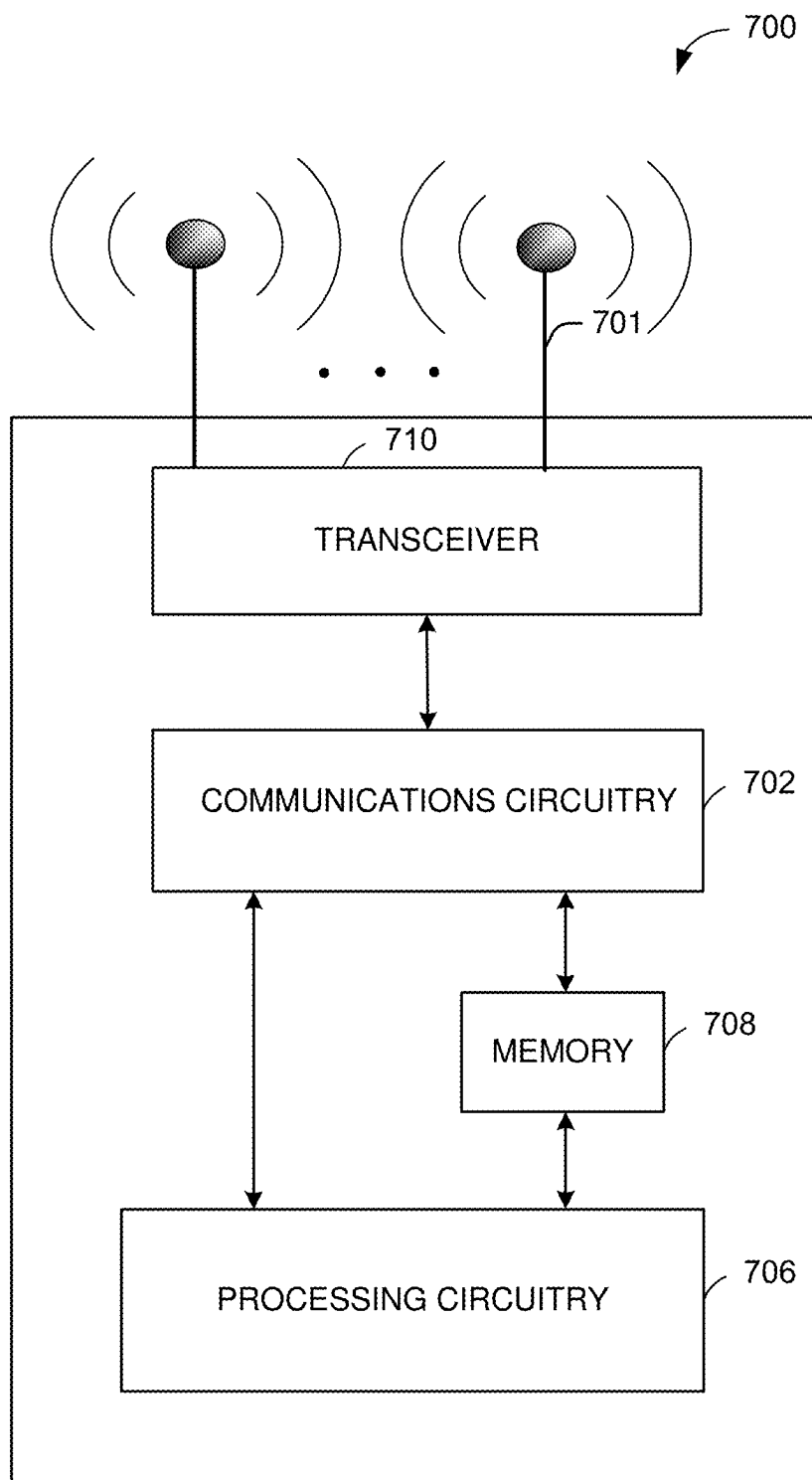


FIG. 7

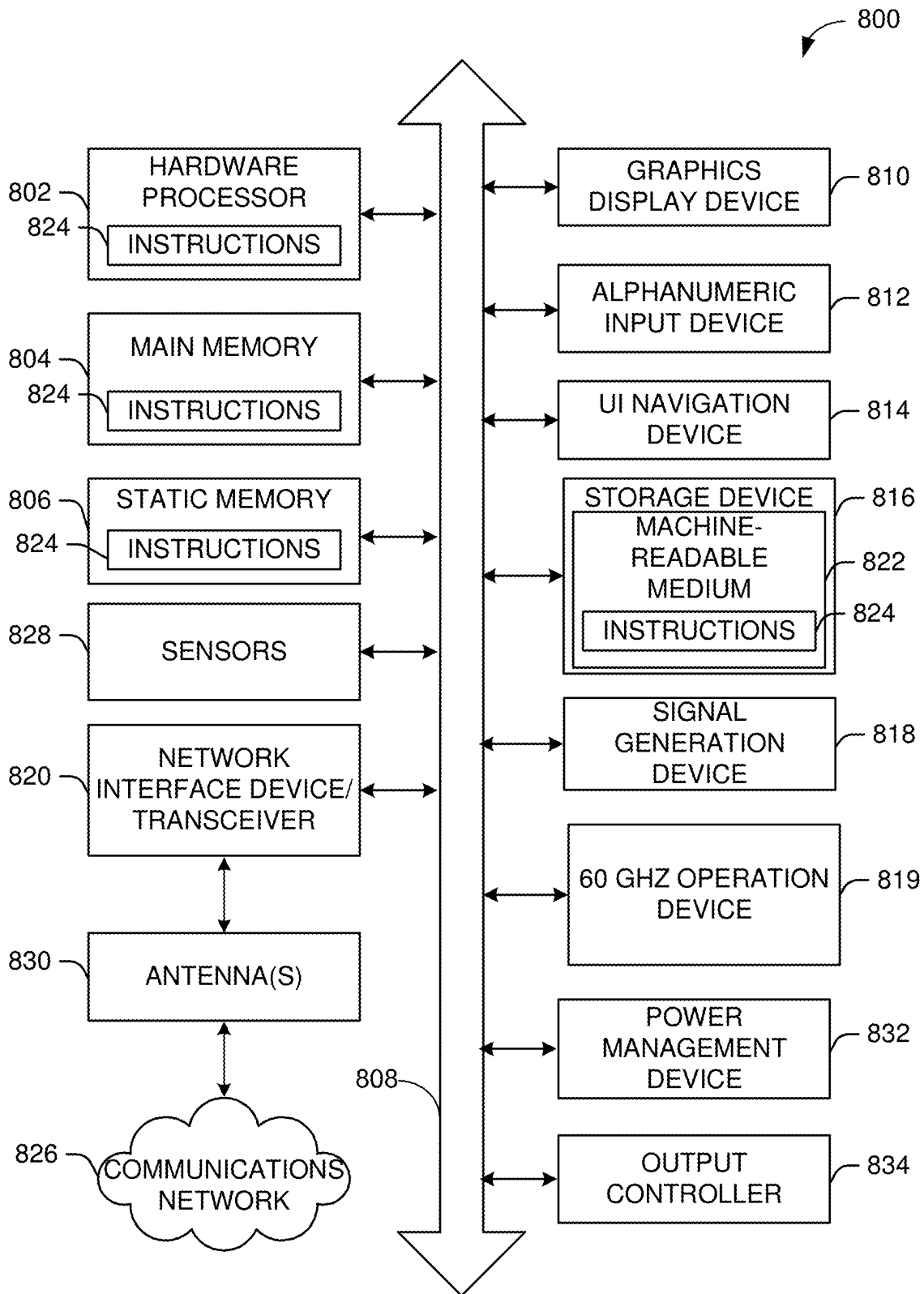


FIG. 8

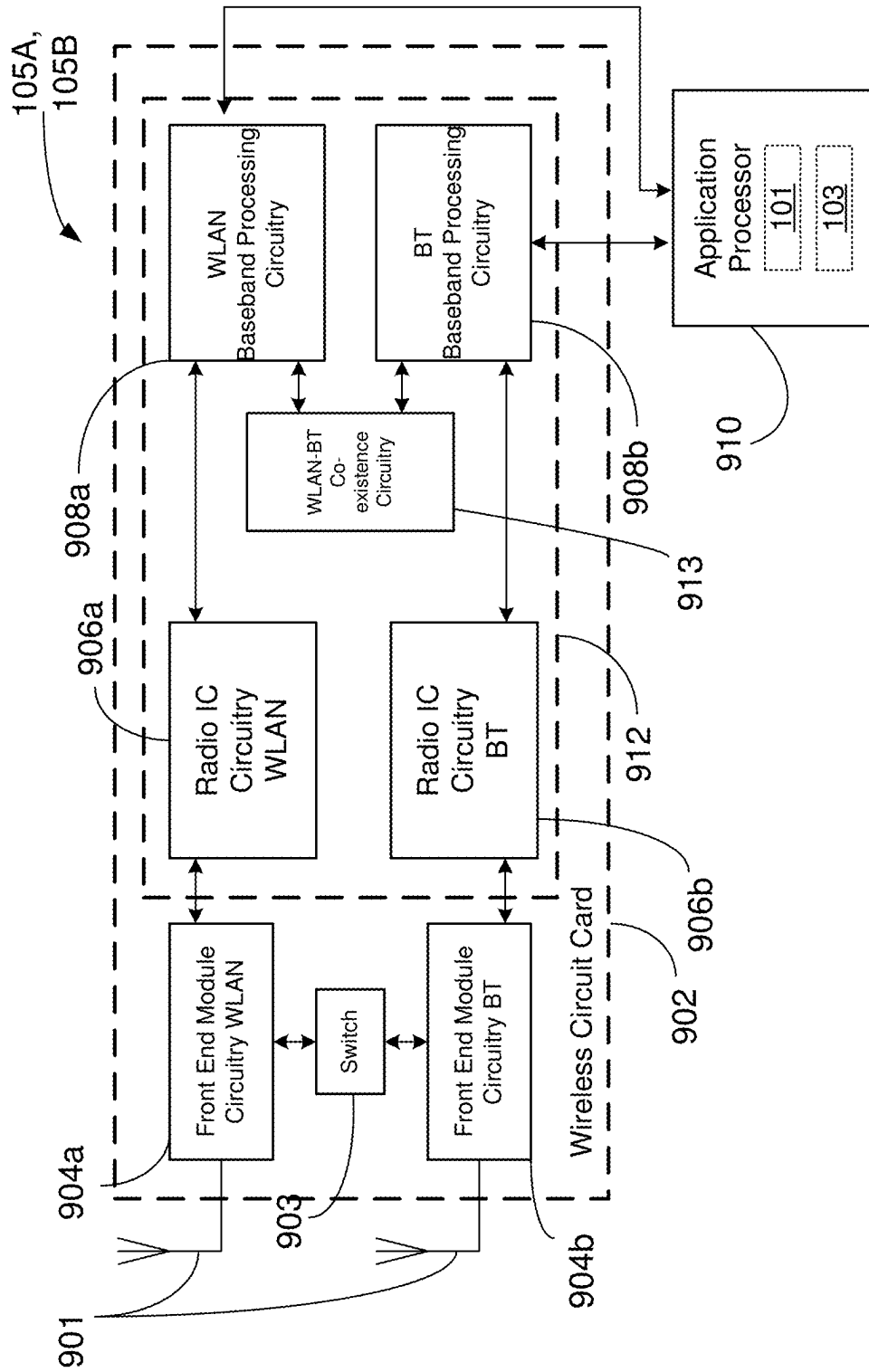


FIG. 9

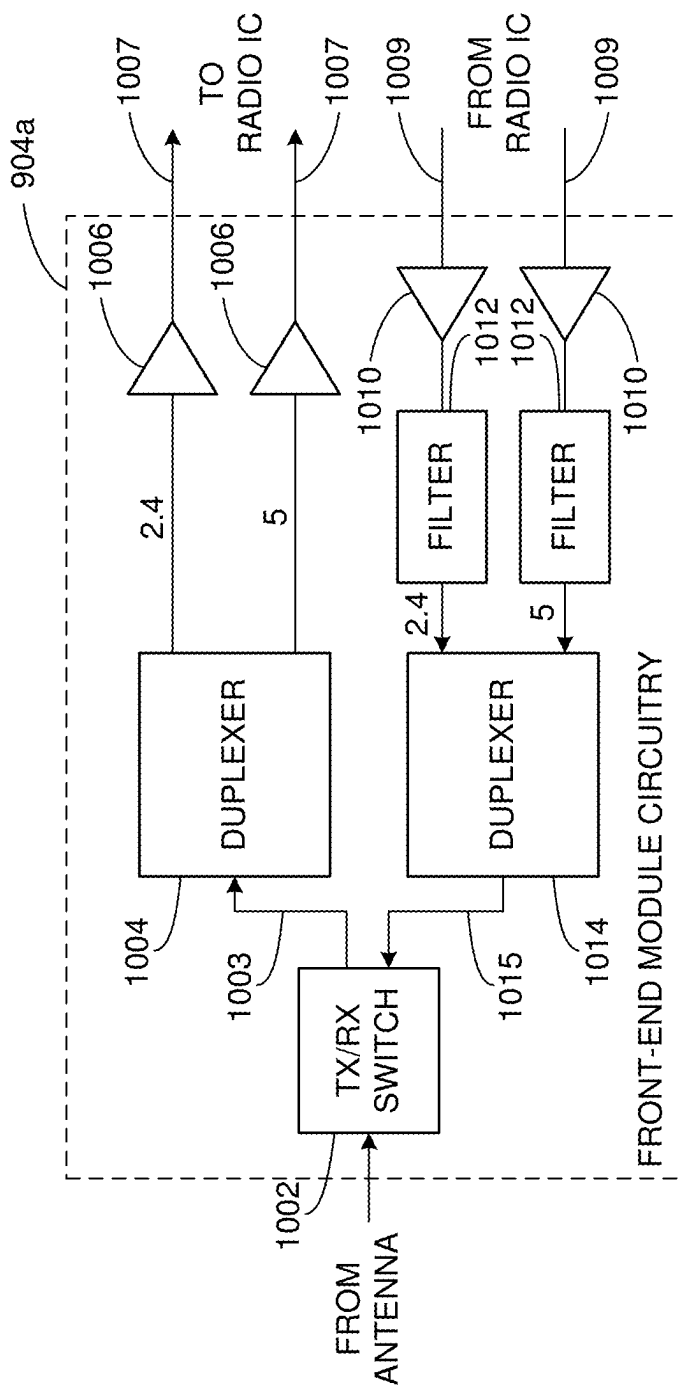


FIG. 10

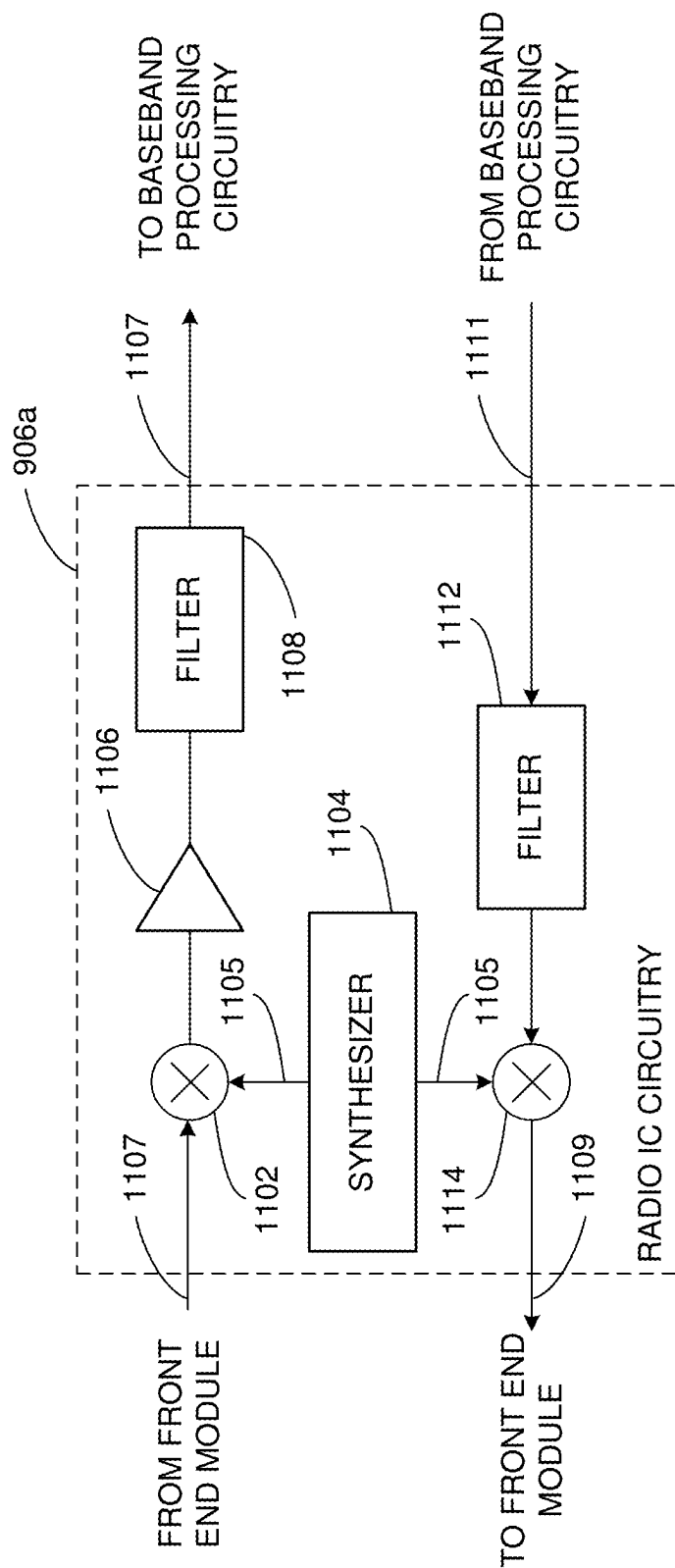


FIG. 11

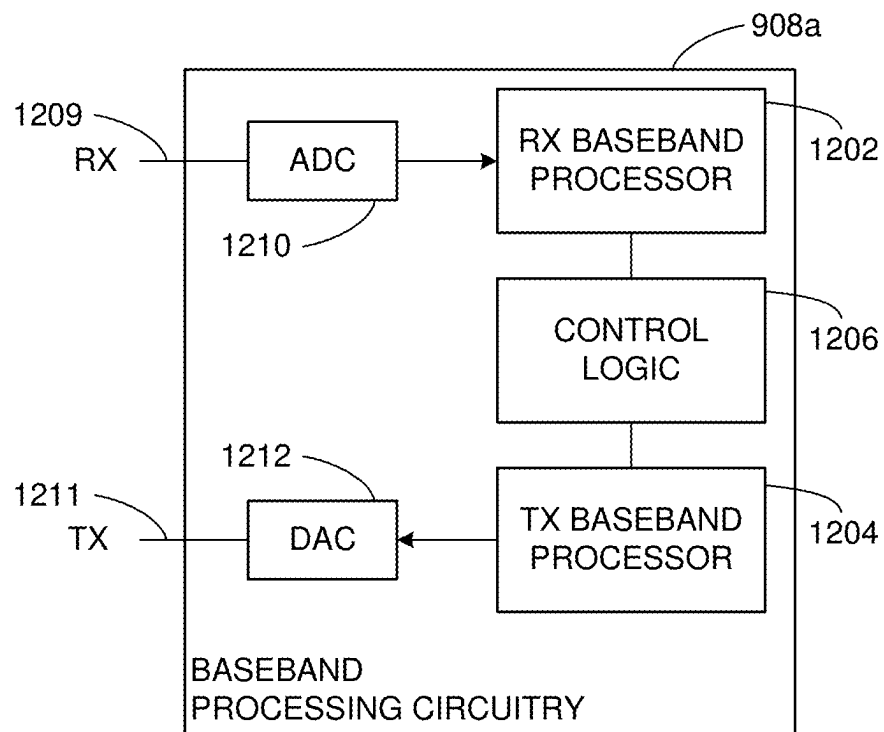


FIG. 12

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NUMEROLOGY FOR 60GHZ OPERATION TO ACHIEVE CHANNELIZATION ALIGNMENT

TECHNICAL FIELD

This disclosure generally relates to systems and methods for wireless communications and, more particularly, to numerology for 60 GHz operation to achieve channelization alignment.

BACKGROUND

Wireless devices are becoming widely prevalent and are increasingly requesting access to wireless channels. The Institute of Electrical and Electronics Engineers (IEEE) is developing one or more standards that utilize Orthogonal Frequency-Division Multiple Access (OFDMA) in channel allocation.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a network diagram illustrating an example network environment for 60 GHz operation, in accordance with one or more example embodiments of the present disclosure.

FIGS. 2-5 depict illustrative schematic diagrams for 60 GHz operation, in accordance with one or more example embodiments of the present disclosure.

FIG. 6 illustrates a flow diagram of a process for an illustrative 60 GHz operation system, in accordance with one or more example embodiments of the present disclosure.

FIG. 7 illustrates a functional diagram of an exemplary communication station that may be suitable for use as a user device, in accordance with one or more example embodiments of the present disclosure.

FIG. 8 illustrates a block diagram of an example machine upon which any of one or more techniques (e.g., methods) may be performed, in accordance with one or more example embodiments of the present disclosure.

FIG. 9 is a block diagram of a radio architecture in accordance with some examples.

FIG. 10 illustrates an example front-end module circuitry for use in the radio architecture of FIG. 9, in accordance with one or more example embodiments of the present disclosure.

FIG. 11 illustrates an example radio IC circuitry for use in the radio architecture of FIG. 9, in accordance with one or more example embodiments of the present disclosure.

FIG. 12 illustrates an example baseband processing circuitry for use in the radio architecture of FIG. 9, in accordance with one or more example embodiments of the present disclosure.

DETAILED DESCRIPTION

The following description and the drawings sufficiently illustrate specific embodiments to enable those skilled in the art to practice them. Other embodiments may incorporate structural, logical, electrical, process, algorithm, and other changes. Portions and features of some embodiments may be included in, or substituted for, those of other embodiments. Embodiments set forth in the claims encompass all available equivalents of those claims.

For Wi-Fi 8, a promising technical direction is to include operation at 60 GHz in mainstream Wi-Fi.

There are several enablers that make this more viable than in the past:

2

Cost reduction that allows reusing as much as possible the same baseband for regular Wi-Fi and 60 GHz radio.

Less potential for throughput enhancements in the lower band.

5 Multi-link framework that makes operation on multiple links easier, and that allows to compensate for the fragility of the link at 60 GHz through easy fall back to lower band operation.

10 A main physical layer (PHY) characteristic for operation at 60 GHz, is to make minimal changes to the baseband design and to reuse most of what was defined in the lower band (playing with upclocking to adjust to the larger bandwidths at 60 GHz and increase subcarrier spacing to fight phase noise. An objective of this disclosure is to define 60 GHz PHY to be as resemblant as possible to the parameters in the lower band.

15 Example embodiments of the present disclosure relate to systems, methods, and devices for numerology for 60 GHz operation to achieve channelization alignment.

20 In one embodiment, a 60 GHz operation system may use the numerology defined for orthogonal frequency-division multiplexing (OFDM) in 802.11ac ("11ac") standard and upclock it in order to increase the bandwidth. The 60 GHz operation system may upclock it by a factor of 6.75 in order to have channelization that is made of fractions of the current channelization for 802.11ad/ay. With the proposal, parameters used for fast Fourier transform (FFT) will be the same (64 subcarriers) for a bandwidth of 135 MHz at 60 GHz as for a bandwidth of 20 MHz in 2.4/5/6 GHz bands. The same is true for 270, 540, and 1080 MHz at 60 GHz (same as bandwidths 40, 80, and 160 MHz in 2.4/5/6 GHz bands). This is shown in FIG. 2.

25 A 60 GHz operation system may facilitate alignment of the new channels with lower bandwidth and the channels used by 11ad/11ay. The above approach reuses the 11ac standard data payload as outlined in that standard as a starting point.

30 Another embodiment is to modify the payload to optimize operation in the new band. Here the proposal is to modify the 11ac preamble or replace it with a new one. While this requires a change to the hardware, the change would be minor compared to a new PHY. The payload would be more or less preserved, the Hardware architecture would be also more or less preserved (FFT sizes, pipeline processing, etc. . . .). This may be required for operation and acceptable performance under the new clocking scheme and at the higher band include.

35 The above descriptions are for purposes of illustration and are not meant to be limiting. Numerous other examples, configurations, processes, algorithms, etc., may exist, some of which are described in greater detail below. Example embodiments will now be described with reference to the accompanying figures.

40 FIG. 1 is a network diagram illustrating an example network environment of 60 GHz operation, according to some example embodiments of the present disclosure. Wireless network 100 may include one or more user devices 120 and one or more access points(s) (AP) 102, which may communicate in accordance with IEEE 802.11 communication standards. The user device(s) 120 may be mobile devices that are non-stationary (e.g., not having fixed locations) or may be stationary devices.

45 In some embodiments, the user devices 120 and the AP 102 may include one or more computer systems similar to that of the functional diagram of FIG. 7 and/or the example machine/system of FIG. 8.

One or more illustrative user device(s) **120** and/or AP(s) **102** may be operable by one or more user(s) **110**. It should be noted that any addressable unit may be a station (STA). An STA may take on multiple distinct characteristics, each of which shape its function. For example, a single addressable unit might simultaneously be a portable STA, a quality-of-service (QoS) STA, a dependent STA, and a hidden STA. The one or more illustrative user device(s) **120** and the AP(s) **102** may be STAs. The one or more illustrative user device(s) **120** and/or AP(s) **102** may operate as a personal basic service set (PBSS) control point/access point (PCP/AP). The user device(s) **120** (e.g., **124**, **126**, or **128**) and/or AP(s) **102** may include any suitable processor-driven device including, but not limited to, a mobile device or a non-mobile, e.g., a static device. For example, user device(s) **120** and/or AP(s) **102** may include, a user equipment (UE), a station (STA), an access point (AP), a software enabled AP (SoftAP), a personal computer (PC), a wearable wireless device (e.g., bracelet, watch, glasses, ring, etc.), a desktop computer, a mobile computer, a laptop computer, an Ultra-book™ computer, a notebook computer, a tablet computer, a server computer, a handheld computer, a handheld device, an internet of things (IoT) device, a sensor device, a PDA device, a handheld PDA device, an on-board device, an off-board device, a hybrid device (e.g., combining cellular phone functionalities with PDA device functionalities), a consumer device, a vehicular device, a non-vehicular device, a mobile or portable device, a non-mobile or non-portable device, a mobile phone, a cellular telephone, a PCS device, a PDA device which incorporates a wireless communication device, a mobile or portable GPS device, a DVB device, a relatively small computing device, a non-desktop computer, a “carry small live large” (CSLL) device, an ultra mobile device (UMD), an ultra mobile PC (UMPC), a mobile internet device (MID), an “origami” device or computing device, a device that supports dynamically composable computing (DCC), a context-aware device, a video device, an audio device, an A/V device, a set-top-box (STB), a blu-ray disc (BD) player, a BD recorder, a digital video disc (DVD) player, a high definition (HD) DVD player, a DVD recorder, a HD DVD recorder, a personal video recorder (PVR), a broadcast HD receiver, a video source, an audio source, a video sink, an audio sink, a stereo tuner, a broadcast radio receiver, a flat panel display, a personal media player (PMP), a digital video camera (DVC), a digital audio player, a speaker, an audio receiver, an audio amplifier, a gaming device, a data source, a data sink, a digital still camera (DSC), a media player, a smartphone, a television, a music player, or the like. Other devices, including smart devices such as lamps, climate control, car components, household components, appliances, etc. may also be included in this list.

As used herein, the term “Internet of Things (IoT) device” is used to refer to any object (e.g., an appliance, a sensor, etc.) that has an addressable interface (e.g., an Internet protocol (IP) address, a Bluetooth identifier (ID), a near-field communication (NFC) ID, etc.) and can transmit information to one or more other devices over a wired or wireless connection. An IoT device may have a passive communication interface, such as a quick response (QR) code, a radio-frequency identification (RFID) tag, an NFC tag, or the like, or an active communication interface, such as a modem, a transceiver, a transmitter-receiver, or the like. An IoT device can have a particular set of attributes (e.g., a device state or status, such as whether the IoT device is on or off, open or closed, idle or active, available for task execution or busy, and so on, a cooling or heating function,

an environmental monitoring or recording function, a light-emitting function, a sound-emitting function, etc.) that can be embedded in and/or controlled/monitored by a central processing unit (CPU), microprocessor, ASIC, or the like, and configured for connection to an IoT network such as a local ad-hoc network or the Internet. For example, IoT devices may include, but are not limited to, refrigerators, toasters, ovens, microwaves, freezers, dishwashers, dishes, hand tools, clothes washers, clothes dryers, furnaces, air conditioners, thermostats, televisions, light fixtures, vacuum cleaners, sprinklers, electricity meters, gas meters, etc., so long as the devices are equipped with an addressable communications interface for communicating with the IoT network. IoT devices may also include cell phones, desktop computers, laptop computers, tablet computers, personal digital assistants (PDAs), etc. Accordingly, the IoT network may be comprised of a combination of “legacy” Internet-accessible devices (e.g., laptop or desktop computers, cell phones, etc.) in addition to devices that do not typically have Internet-connectivity (e.g., dishwashers, etc.).

The user device(s) **120** and/or AP(s) **102** may also include mesh stations in, for example, a mesh network, in accordance with one or more IEEE 802.11 standards and/or 3GPP standards.

Any of the user device(s) **120** (e.g., user devices **124**, **126**, **128**), and AP(s) **102** may be configured to communicate with each other via one or more communications networks **130** and/or **135** wirelessly or wired. The user device(s) **120** may also communicate peer-to-peer or directly with each other with or without the AP(s) **102**. Any of the communications networks **130** and/or **135** may include, but not limited to, any one of a combination of different types of suitable communications networks such as, for example, broadcasting networks, cable networks, public networks (e.g., the Internet), private networks, wireless networks, cellular networks, or any other suitable private and/or public networks. Further, any of the communications networks **130** and/or **135** may have any suitable communication range associated therewith and may include, for example, global networks (e.g., the Internet), metropolitan area networks (MANs), wide area networks (WANs), local area networks (LANs), or personal area networks (PANs). In addition, any of the communications networks **130** and/or **135** may include any type of medium over which network traffic may be carried including, but not limited to, coaxial cable, twisted-pair wire, optical fiber, a hybrid fiber coaxial (HFC) medium, microwave terrestrial transceivers, radio frequency communication mediums, white space communication mediums, ultra-high frequency communication mediums, satellite communication mediums, or any combination thereof.

Any of the user device(s) **120** (e.g., user devices **124**, **126**, **128**) and AP(s) **102** may include one or more communications antennas. The one or more communications antennas may be any suitable type of antennas corresponding to the communications protocols used by the user device(s) **120** (e.g., user devices **124**, **126** and **128**), and AP(s) **102**. Some non-limiting examples of suitable communications antennas include Wi-Fi antennas, Institute of Electrical and Electronics Engineers (IEEE) 802.11 family of standards compatible antennas, directional antennas, non-directional antennas, dipole antennas, folded dipole antennas, patch antennas, multiple-input multiple-output (MIMO) antennas, omnidirectional antennas, quasi-omnidirectional antennas, or the like. The one or more communications antennas may be communicatively coupled to a radio component to transmit and/or receive signals, such as communications signals to and/or from the user devices **120** and/or AP(s) **102**.

Any of the user device(s) 120 (e.g., user devices 124, 126, 128), and AP(s) 102 may be configured to perform directional transmission and/or directional reception in conjunction with wirelessly communicating in a wireless network. Any of the user device(s) 120 (e.g., user devices 124, 126, 128), and AP(s) 102 may be configured to perform such directional transmission and/or reception using a set of multiple antenna arrays (e.g., DMG antenna arrays or the like). Each of the multiple antenna arrays may be used for transmission and/or reception in a particular respective direction or range of directions. Any of the user device(s) 120 (e.g., user devices 124, 126, 128), and AP(s) 102 may be configured to perform any given directional transmission towards one or more defined transmit sectors. Any of the user device(s) 120 (e.g., user devices 124, 126, 128), and AP(s) 102 may be configured to perform any given directional reception from one or more defined receive sectors.

MIMO beamforming in a wireless network may be accomplished using RF beamforming and/or digital beamforming. In some embodiments, in performing a given MIMO transmission, user devices 120 and/or AP(s) 102 may be configured to use all or a subset of its one or more communications antennas to perform MIMO beamforming.

Any of the user devices 120 (e.g., user devices 124, 126, 128), and AP(s) 102 may include any suitable radio and/or transceiver for transmitting and/or receiving radio frequency (RF) signals in the bandwidth and/or channels corresponding to the communications protocols utilized by any of the user device(s) 120 and AP(s) 102 to communicate with each other. The radio components may include hardware and/or software to modulate and/or demodulate communications signals according to pre-established transmission protocols. The radio components may further have hardware and/or software instructions to communicate via one or more Wi-Fi and/or Wi-Fi direct protocols, as standardized by the Institute of Electrical and Electronics Engineers (IEEE) 802.11 standards. In certain example embodiments, the radio component, in cooperation with the communications antennas, may be configured to communicate via 2.4 GHz channels (e.g. 802.11b, 802.11g, 802.11n, 802.11ax), 5 GHz channels (e.g. 802.11n, 802.11ac, 802.11ax, 802.11be, etc.), 6 GHz channels (e.g., 802.11ax, 802.11be, etc.), or 60 GHz channels (e.g. 802.11ad, 802.11ay). 800 MHz channels (e.g. 802.11ah). The communications antennas may operate at 28 GHz and 40 GHz. It should be understood that this list of communication channels in accordance with certain 802.11 standards is only a partial list and that other 802.11 standards may be used (e.g., Next Generation Wi-Fi, or other standards). In some embodiments, non-Wi-Fi protocols may be used for communications between devices, such as Bluetooth, dedicated short-range communication (DSRC), Ultra-High Frequency (UHF) (e.g. IEEE 802.11af, IEEE 802.22), white band frequency (e.g., white spaces), or other packetized radio communications. The radio component may include any known receiver and baseband suitable for communicating via the communications protocols. The radio component may further include a low noise amplifier (LNA), additional signal amplifiers, an analog-to-digital (A/D) converter, one or more buffers, and digital baseband.

In one embodiment, and with reference to FIG. 1, a user device 120 may be in communication with one or more APs 102. For example, one or more APs 102 may implement a 60 GHz operation 142 with one or more user devices 120. It is understood that the above descriptions are for purposes of illustration and are not meant to be limiting.

It is understood that the above descriptions are for purposes of illustration and are not meant to be limiting.

FIGS. 2-5 depict illustrative schematic diagrams for 60 GHz operation, in accordance with one or more example embodiments of the present disclosure.

Referring to FIG. 2, there is shown the numerology with the approach associated with 60 GHz operation/communication.

Table 202 is for 11ac legacy numerology. N_{DFT} is number of symbols per block symbol at the baseband level. As can be seen that N_{DFT} varies for each bandwidth (e.g., 20, 40, 80, 160 MHz in Table 202). However, in Table 204, N_{DFT} is selected to be the same for the various bandwidths (e.g., 135 (which is equal to 20 MHz $\times$ 6.75), 270 (which is equal to 40 MHz $\times$ 6.75), 540 (which is equal to 80 MHz $\times$ 6.75), 1080 (which is equal to 160 MHz $\times$ 6.75) MHz) in the 60 GHz frequency.

Δf is subcarrier spacing between each carrier/frequency. For example, for a 135 MHz, 270 MHz, 540 MHz, 1080 MHz channels at 60 GHz the subcarrier spacing may be kept the same at 2.11 MHz. This value may be determined based on upclocking the 11ac subcarrier spacing of 312.5 KHz by 6.75 $\times$ for the listed frequencies.

For T_{DFT} (time length of an OFDM symbol) for a 135 MHz, 270 MHz, 540 MHz, 1080 MHz channels at 60 GHz the T_{DFT} may be 0.4 μ s. This value is determined based on upclocking (in this case divided) the 11ac T_{DFT} of 3.2 μ s by 6.75 for the 135 MHz, 270 MHz, 540 MHz, 1080 MHz. For T_s , which is cadence in time or subcarrier spacing in time, the values are determined by taking the 20 MHz, 40 MHz, 80 MHz, and 160 MHz numbers in the legacy Table 202 (50, 25, 12.5, 6.25 ns, respectively) and dividing by 6.75 $\times$ for frequencies 135 MHz, 270 MHz, 540 MHz, 1080 MHz, respectively.

Proposed approaches are listed below. These are just general changes, the details are left to further design and a later proposal.

Changes in the 11ac Legacy and very high throughput (VHT) short training field (STF) structure. This may include repeated STF structures. This portion of the payload could be expanded to allow beamforming training during this period.

There may be modifications to the long training field (LTF) structure to improve receiver parameter estimation. Potentially this would be for both the Legacy and the VHT portion.

The Legacy and VHT STF/LTF could be modified as a full entity to create one longer training period for use in this band.

Modification to the Legacy or VHT signal field. This would allow parameters relevant to this band, along with any changes to the STF/LTF to be signaled

Definition of a new signal field and removal of the Legacy and VHT Signal fields.

Definition of a new Green field preamble for the VHT payload. Since there would be no legacy devices, the overhead of signaling that is done in 11ac (VHT) is not required, and this duration of time in the packet could be used to build an optimized preamble. Basically, the preamble would be replaced with a new preamble appropriate for 60 GHz operation (Omni detection, beam selection/training, parameter estimation, etc). The signal fields would be optimized, with the removal of the legacy SIG which is not needed, and a new VHT SIG.

In one or more embodiments, a 60 GHz operation system may allow to fully reuse the baseband used in the lower band, with a few changes as possible. Only adjust the RF parameters in order to be able to use channelization that will match with the channelization used in 11ad/11ay, and claim

better coexistence in the sense the energy detection will be able to be done on overlapping channels.

FIG. 6 illustrates a flow diagram of illustrative process 600 for a 60 GHz operation system, in accordance with one or more example embodiments of the present disclosure.

At block 602, a device (e.g., the user device(s) 120 and/or the AP 102 of FIG. 1 and/or the 60 GHz operation device 819 of FIG. 8) may utilize a legacy baseband data for a generation of a plurality of orthogonal frequency-division multiplexing (OFDM) symbols for a 60 GHz communication.

At block 604, the device may select a number of subcarriers based on a bandwidth selection. The number of subcarriers is at least one of 64, 128, 256, or 512. The legacy baseband data is associated with a 20 MHz frequency for IEEE 802.11ac. The legacy subcarriers spacing is 312.5 kHz. The upclocking factor is 6.75.

At block 606, the device may generate an upclock of the legacy baseband data by an upclocking factor that aligns a first bandwidth with a legacy bandwidth.

At block 608, the device may assign a first subcarrier spacing for a first group of bandwidths.

At block 610, the device may assign a second subcarrier spacing for a second group of bandwidths.

At block 612, the device may generate the plurality of OFDM symbols using an upclocking mechanism based on a bandwidth for the 60 GHz communication.

At block 614, the device may cause to send a plurality of OFDM symbols to a station device.

In one or more embodiments, the device may upclock the legacy baseband data by 6.75 times for a 135 MHz bandwidth in the 60 GHz communication. The device may utilize a first legacy data for a 40 MHz physical layer (PHY) convergence protocol data unit (PPDU). The device may upclock the first legacy data by 6.75 times for a 270 MHz bandwidth for the 60 GHz communication. The device may utilize a first legacy data for an 80 MHz physical layer (PHY) convergence protocol data unit (PPDU). The device may upclock the first legacy data by 6.75 times for a 540 MHz bandwidth for the 60 GHz communication. The device may utilize a first legacy data for a 160 MHz physical layer (PHY) convergence protocol data unit (PPDU). The device may upclock the first legacy data by 6.75 times for a 1080 MHz bandwidth for the 60 GHz communication.

It is understood that the above descriptions are for purposes of illustration and are not meant to be limiting.

FIG. 7 shows a functional diagram of an exemplary communication station 700, in accordance with one or more example embodiments of the present disclosure. In one embodiment, FIG. 7 illustrates a functional block diagram of a communication station that may be suitable for use as an AP 102 (FIG. 1) or a user device 120 (FIG. 1) in accordance with some embodiments. The communication station 700 may also be suitable for use as a handheld device, a mobile device, a cellular telephone, a smartphone, a tablet, a notebook, a wireless terminal, a laptop computer, a wearable computer device, a femtocell, a high data rate (HDR) subscriber station, an access point, an access terminal, or other personal communication system (PCS) device.

The communication station 700 may include communications circuitry 702 and a transceiver 710 for transmitting and receiving signals to and from other communication stations using one or more antennas 701. The communications circuitry 702 may include circuitry that can operate the physical layer (PHY) communications and/or medium access control (MAC) communications for controlling access to the wireless medium, and/or any other communi-

cations layers for transmitting and receiving signals. The communication station 700 may also include processing circuitry 706 and memory 708 arranged to perform the operations described herein. In some embodiments, the communications circuitry 702 and the processing circuitry 706 may be configured to perform operations detailed in the above figures, diagrams, and flows.

In accordance with some embodiments, the communications circuitry 702 may be arranged to contend for a wireless medium and configure frames or packets for communicating over the wireless medium. The communications circuitry 702 may be arranged to transmit and receive signals. The communications circuitry 702 may also include circuitry for modulation/demodulation, upconversion/downconversion, filtering, amplification, etc. In some embodiments, the processing circuitry 706 of the communication station 700 may include one or more processors. In other embodiments, two or more antennas 701 may be coupled to the communications circuitry 702 arranged for sending and receiving signals. The memory 708 may store information for configuring the processing circuitry 706 to perform operations for configuring and transmitting message frames and performing the various operations described herein. The memory 708 may include any type of memory, including non-transitory memory, for storing information in a form readable by a machine (e.g., a computer). For example, the memory 708 may include a computer-readable storage device, read-only memory (ROM), random-access memory (RAM), magnetic disk storage media, optical storage media, flash-memory devices and other storage devices and media.

In some embodiments, the communication station 700 may be part of a portable wireless communication device, such as a personal digital assistant (PDA), a laptop or portable computer with wireless communication capability, a web tablet, a wireless telephone, a smartphone, a wireless headset, a pager, an instant messaging device, a digital camera, an access point, a television, a medical device (e.g., a heart rate monitor, a blood pressure monitor, etc.), a wearable computer device, or another device that may receive and/or transmit information wirelessly.

In some embodiments, the communication station 700 may include one or more antennas 701. The antennas 701 may include one or more directional or omnidirectional antennas, including, for example, dipole antennas, monopole antennas, patch antennas, loop antennas, microstrip antennas, or other types of antennas suitable for transmission of RF signals. In some embodiments, instead of two or more antennas, a single antenna with multiple apertures may be used. In these embodiments, each aperture may be considered a separate antenna. In some multiple-input multiple-output (MIMO) embodiments, the antennas may be effectively separated for spatial diversity and the different channel characteristics that may result between each of the antennas and the antennas of a transmitting station.

In some embodiments, the communication station 700 may include one or more of a keyboard, a display, a non-volatile memory port, multiple antennas, a graphics processor, an application processor, speakers, and other mobile device elements. The display may be an LCD screen including a touch screen.

Although the communication station 700 is illustrated as having several separate functional elements, two or more of the functional elements may be combined and may be implemented by combinations of software-configured elements, such as processing elements including digital signal processors (DSPs), and/or other hardware elements. For example, some elements may include one or more micro-

processors, DSPs, field-programmable gate arrays (FPGAs), application specific integrated circuits (ASICs), radio-frequency integrated circuits (RFICs) and combinations of various hardware and logic circuitry for performing at least the functions described herein. In some embodiments, the functional elements of the communication station **700** may refer to one or more processes operating on one or more processing elements.

Certain embodiments may be implemented in one or a combination of hardware, firmware, and software. Other embodiments may also be implemented as instructions stored on a computer-readable storage device, which may be read and executed by at least one processor to perform the operations described herein. A computer-readable storage device may include any non-transitory memory mechanism for storing information in a form readable by a machine (e.g., a computer). For example, a computer-readable storage device may include read-only memory (ROM), random-access memory (RAM), magnetic disk storage media, optical storage media, flash-memory devices, and other storage devices and media. In some embodiments, the communication station **700** may include one or more processors and may be configured with instructions stored on a computer-readable storage device.

FIG. **8** illustrates a block diagram of an example of a machine **800** or system upon which any one or more of the techniques (e.g., methodologies) discussed herein may be performed. In other embodiments, the machine **800** may operate as a standalone device or may be connected (e.g., networked) to other machines. In a networked deployment, the machine **800** may operate in the capacity of a server machine, a client machine, or both in server-client network environments. In an example, the machine **800** may act as a peer machine in peer-to-peer (P2P) (or other distributed) network environments. The machine **800** may be a personal computer (PC), a tablet PC, a set-top box (STB), a personal digital assistant (PDA), a mobile telephone, a wearable computer device, a web appliance, a network router, a switch or bridge, or any machine capable of executing instructions (sequential or otherwise) that specify actions to be taken by that machine, such as a base station. Further, while only a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein, such as cloud computing, software as a service (SaaS), or other computer cluster configurations.

Examples, as described herein, may include or may operate on logic or a number of components, modules, or mechanisms. Modules are tangible entities (e.g., hardware) capable of performing specified operations when operating. A module includes hardware. In an example, the hardware may be specifically configured to carry out a specific operation (e.g., hardwired). In another example, the hardware may include configurable execution units (e.g., transistors, circuits, etc.) and a computer readable medium containing instructions where the instructions configure the execution units to carry out a specific operation when in operation. The configuring may occur under the direction of the executions units or a loading mechanism. Accordingly, the execution units are communicatively coupled to the computer-readable medium when the device is operating. In this example, the execution units may be a member of more than one module. For example, under operation, the execution units may be configured by a first set of instructions to implement a first

module at one point in time and reconfigured by a second set of instructions to implement a second module at a second point in time.

The machine (e.g., computer system) **800** may include a hardware processor **802** (e.g., a central processing unit (CPU), a graphics processing unit (GPU), a hardware processor core, or any combination thereof), a main memory **804** and a static memory **806**, some or all of which may communicate with each other via an interlink (e.g., bus) **808**. The machine **800** may further include a power management device **832**, a graphics display device **810**, an alphanumeric input device **812** (e.g., a keyboard), and a user interface (UI) navigation device **814** (e.g., a mouse). In an example, the graphics display device **810**, alphanumeric input device **812**, and UI navigation device **814** may be a touch screen display. The machine **800** may additionally include a storage device (i.e., drive unit) **816**, a signal generation device **818** (e.g., a speaker), a 60 GHz operation device **819**, a network interface device/transceiver **820** coupled to antenna(s) **830**, and one or more sensors **828**, such as a global positioning system (GPS) sensor, a compass, an accelerometer, or other sensor. The machine **800** may include an output controller **834**, such as a serial (e.g., universal serial bus (USB), parallel, or other wired or wireless (e.g., infrared (IR), near field communication (NFC), etc.) connection to communicate with or control one or more peripheral devices (e.g., a printer, a card reader, etc.)). The operations in accordance with one or more example embodiments of the present disclosure may be carried out by a baseband processor. The baseband processor may be configured to generate corresponding baseband signals. The baseband processor may further include physical layer (PHY) and medium access control layer (MAC) circuitry, and may further interface with the hardware processor **802** for generation and processing of the baseband signals and for controlling operations of the main memory **804**, the storage device **816**, and/or the 60 GHz operation device **819**. The baseband processor may be provided on a single radio card, a single chip, or an integrated circuit (IC).

The storage device **816** may include a machine readable medium **822** on which is stored one or more sets of data structures or instructions **824** (e.g., software) embodying or utilized by any one or more of the techniques or functions described herein. The instructions **824** may also reside, completely or at least partially, within the main memory **804**, within the static memory **806**, or within the hardware processor **802** during execution thereof by the machine **800**. In an example, one or any combination of the hardware processor **802**, the main memory **804**, the static memory **806**, or the storage device **816** may constitute machine-readable media.

The 60 GHz operation device **819** may carry out or perform any of the operations and processes (e.g., process **600**) described and shown above.

It is understood that the above are only a subset of what the 60 GHz operation device **819** may be configured to perform and that other functions included throughout this disclosure may also be performed by the 60 GHz operation device **819**.

While the machine-readable medium **822** is illustrated as a single medium, the term “machine-readable medium” may include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) configured to store the one or more instructions **824**.

Various embodiments may be implemented fully or partially in software and/or firmware. This software and/or firmware may take the form of instructions contained in or on a non-transitory computer-readable storage medium.

Those instructions may then be read and executed by one or more processors to enable performance of the operations described herein. The instructions may be in any suitable form, such as but not limited to source code, compiled code, interpreted code, executable code, static code, dynamic code, and the like. Such a computer-readable medium may include any tangible non-transitory medium for storing information in a form readable by one or more computers, such as but not limited to read only memory (ROM); random access memory (RAM); magnetic disk storage media; optical storage media; a flash memory, etc.

The term “machine-readable medium” may include any medium that is capable of storing, encoding, or carrying instructions for execution by the machine **800** and that cause the machine **800** to perform any one or more of the techniques of the present disclosure, or that is capable of storing, encoding, or carrying data structures used by or associated with such instructions. Non-limiting machine-readable medium examples may include solid-state memories and optical and magnetic media. In an example, a massed machine-readable medium includes a machine-readable medium with a plurality of particles having resting mass. Specific examples of massed machine-readable media may include non-volatile memory, such as semiconductor memory devices (e.g., electrically programmable read-only memory (EPROM), or electrically erasable programmable read-only memory (EEPROM)) and flash memory devices; magnetic disks, such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks.

The instructions **824** may further be transmitted or received over a communications network **826** using a transmission medium via the network interface device/transceiver **820** utilizing any one of a number of transfer protocols (e.g., frame relay, internet protocol (IP), transmission control protocol (TCP), user datagram protocol (UDP), hypertext transfer protocol (HTTP), etc.). Example communications networks may include a local area network (LAN), a wide area network (WAN), a packet data network (e.g., the Internet), mobile telephone networks (e.g., cellular networks), plain old telephone (POTS) networks, wireless data networks (e.g., Institute of Electrical and Electronics Engineers (IEEE) 802.11 family of standards known as Wi-Fi®, IEEE 802.16 family of standards known as WiMax®, IEEE 802.15.4 family of standards, and peer-to-peer (P2P) networks, among others. In an example, the network interface device/transceiver **820** may include one or more physical jacks (e.g., Ethernet, coaxial, or phone jacks) or one or more antennas to connect to the communications network **826**. In an example, the network interface device/transceiver **820** may include a plurality of antennas to wirelessly communicate using at least one of single-input multiple-output (SIMO), multiple-input multiple-output (MIMO), or multiple-input single-output (MISO) techniques. The term “transmission medium” shall be taken to include any intangible medium that is capable of storing, encoding, or carrying instructions for execution by the machine **800** and includes digital or analog communications signals or other intangible media to facilitate communication of such software.

The operations and processes described and shown above may be carried out or performed in any suitable order as desired in various implementations. Additionally, in certain implementations, at least a portion of the operations may be carried out in parallel. Furthermore, in certain implementations, less than or more than the operations described may be performed.

FIG. **9** is a block diagram of a radio architecture **105A**, **105B** in accordance with some embodiments that may be implemented in any one of the example APs **102** and/or the example STAs **120** of FIG. **1**. Radio architecture **105A**, **105B** may include radio front-end module (FEM) circuitry **904a-b**, radio IC circuitry **906a-b** and baseband processing circuitry **908a-b**. Radio architecture **105A**, **105B** as shown includes both Wireless Local Area Network (WLAN) functionality and Bluetooth (BT) functionality although embodiments are not so limited. In this disclosure, “WLAN” and “Wi-Fi” are used interchangeably.

FEM circuitry **904a-b** may include a WLAN or Wi-Fi FEM circuitry **904a** and a Bluetooth (BT) FEM circuitry **904b**. The WLAN FEM circuitry **904a** may include a receive signal path comprising circuitry configured to operate on WLAN RF signals received from one or more antennas **901**, to amplify the received signals and to provide the amplified versions of the received signals to the WLAN radio IC circuitry **906a** for further processing. The BT FEM circuitry **904b** may include a receive signal path which may include circuitry configured to operate on BT RF signals received from one or more antennas **901**, to amplify the received signals and to provide the amplified versions of the received signals to the BT radio IC circuitry **906b** for further processing. FEM circuitry **904a** may also include a transmit signal path which may include circuitry configured to amplify WLAN signals provided by the radio IC circuitry **906a** for wireless transmission by one or more of the antennas **901**. In addition, FEM circuitry **904b** may also include a transmit signal path which may include circuitry configured to amplify BT signals provided by the radio IC circuitry **906b** for wireless transmission by the one or more antennas. In the embodiment of FIG. **9**, although FEM **904a** and FEM **904b** are shown as being distinct from one another, embodiments are not so limited, and include within their scope the use of an FEM (not shown) that includes a transmit path and/or a receive path for both WLAN and BT signals, or the use of one or more FEM circuitries where at least some of the FEM circuitries share transmit and/or receive signal paths for both WLAN and BT signals.

Radio IC circuitry **906a-b** as shown may include WLAN radio IC circuitry **906a** and BT radio IC circuitry **906b**. The WLAN radio IC circuitry **906a** may include a receive signal path which may include circuitry to down-convert WLAN RF signals received from the FEM circuitry **904a** and provide baseband signals to WLAN baseband processing circuitry **908a**. BT radio IC circuitry **906b** may in turn include a receive signal path which may include circuitry to down-convert BT RF signals received from the FEM circuitry **904b** and provide baseband signals to BT baseband processing circuitry **908b**. WLAN radio IC circuitry **906a** may also include a transmit signal path which may include circuitry to up-convert WLAN baseband signals provided by the WLAN baseband processing circuitry **908a** and provide WLAN RF output signals to the FEM circuitry **904a** for subsequent wireless transmission by the one or more antennas **901**. BT radio IC circuitry **906b** may also include a transmit signal path which may include circuitry to up-convert BT baseband signals provided by the BT baseband processing circuitry **908b** and provide BT RF output signals to the FEM circuitry **904b** for subsequent wireless transmission by the one or more antennas **901**. In the embodiment of FIG. **9**, although radio IC circuitries **906a** and **906b** are shown as being distinct from one another, embodiments are not so limited, and include within their scope the use of a radio IC circuitry (not shown) that includes a transmit signal path and/or a receive signal path for both WLAN and BT

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signals, or the use of one or more radio IC circuitries where at least some of the radio IC circuitries share transmit and/or receive signal paths for both WLAN and BT signals.

Baseband processing circuitry **908a-b** may include a WLAN baseband processing circuitry **908a** and a BT baseband processing circuitry **908b**. The WLAN baseband processing circuitry **908a** may include a memory, such as, for example, a set of RAM arrays in a Fast Fourier Transform or Inverse Fast Fourier Transform block (not shown) of the WLAN baseband processing circuitry **908a**. Each of the WLAN baseband circuitry **908a** and the BT baseband circuitry **908b** may further include one or more processors and control logic to process the signals received from the corresponding WLAN or BT receive signal path of the radio IC circuitry **906a-b**, and to also generate corresponding WLAN or BT baseband signals for the transmit signal path of the radio IC circuitry **906a-b**. Each of the baseband processing circuitries **908a** and **908b** may further include physical layer (PHY) and medium access control layer (MAC) circuitry, and may further interface with a device for generation and processing of the baseband signals and for controlling operations of the radio IC circuitry **906a-b**.

Referring still to FIG. 9, according to the shown embodiment, WLAN-BT coexistence circuitry **913** may include logic providing an interface between the WLAN baseband circuitry **908a** and the BT baseband circuitry **908b** to enable use cases requiring WLAN and BT coexistence. In addition, a switch **903** may be provided between the WLAN FEM circuitry **904a** and the BT FEM circuitry **904b** to allow switching between the WLAN and BT radios according to application needs. In addition, although the antennas **901** are depicted as being respectively connected to the WLAN FEM circuitry **904a** and the BT FEM circuitry **904b**, embodiments include within their scope the sharing of one or more antennas as between the WLAN and BT FEMs, or the provision of more than one antenna connected to each of FEM **904a** or **904b**.

In some embodiments, the front-end module circuitry **904a-b**, the radio IC circuitry **906a-b**, and baseband processing circuitry **908a-b** may be provided on a single radio card, such as wireless radio card **902**. In some other embodiments, the one or more antennas **901**, the FEM circuitry **904a-b** and the radio IC circuitry **906a-b** may be provided on a single radio card. In some other embodiments, the radio IC circuitry **906a-b** and the baseband processing circuitry **908a-b** may be provided on a single chip or integrated circuit (IC), such as IC **912**.

In some embodiments, the wireless radio card **902** may include a WLAN radio card and may be configured for Wi-Fi communications, although the scope of the embodiments is not limited in this respect. In some of these embodiments, the radio architecture **105A**, **105B** may be configured to receive and transmit orthogonal frequency division multiplexed (OFDM) or orthogonal frequency division multiple access (OFDMA) communication signals over a multicarrier communication channel. The OFDM or OFDMA signals may comprise a plurality of orthogonal subcarriers.

In some of these multicarrier embodiments, radio architecture **105A**, **105B** may be part of a Wi-Fi communication station (STA) such as a wireless access point (AP), a base station or a mobile device including a Wi-Fi device. In some of these embodiments, radio architecture **105A**, **105B** may be configured to transmit and receive signals in accordance with specific communication standards and/or protocols, such as any of the Institute of Electrical and Electronics Engineers (IEEE) standards including, 802.11n-2009, IEEE

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802.11-2012, IEEE 802.11-2016, 802.11n-2009, 802.11ac, 802.11ah, 802.11ad, 802.11 ay and/or 802.11ax standards and/or proposed specifications for WLANs, although the scope of embodiments is not limited in this respect. Radio architecture **105A**, **105B** may also be suitable to transmit and/or receive communications in accordance with other techniques and standards.

In some embodiments, the radio architecture **105A**, **105B** may be configured for high-efficiency Wi-Fi (HEW) communications in accordance with the IEEE 802.11ax standard. In these embodiments, the radio architecture **105A**, **105B** may be configured to communicate in accordance with an OFDMA technique, although the scope of the embodiments is not limited in this respect.

In some other embodiments, the radio architecture **105A**, **105B** may be configured to transmit and receive signals transmitted using one or more other modulation techniques such as spread spectrum modulation (e.g., direct sequence code division multiple access (DS-CDMA) and/or frequency hopping code division multiple access (FH-CDMA)), time-division multiplexing (TDM) modulation, and/or frequency-division multiplexing (FDM) modulation, although the scope of the embodiments is not limited in this respect.

In some embodiments, as further shown in FIG. 6, the BT baseband circuitry **908b** may be compliant with a Bluetooth (BT) connectivity standard such as Bluetooth, Bluetooth 8.0 or Bluetooth 6.0, or any other iteration of the Bluetooth Standard.

In some embodiments, the radio architecture **105A**, **105B** may include other radio cards, such as a cellular radio card configured for cellular (e.g., 5GPP such as LTE, LTE-Advanced or 7G communications).

In some IEEE 802.11 embodiments, the radio architecture **105A**, **105B** may be configured for communication over various channel bandwidths including bandwidths having center frequencies of about 900 MHz, 2.4 GHz, 5 GHz, and bandwidths of about 2 MHz, 4 MHz, 5 MHz, 5.5 MHz, 6 MHz, 8 MHz, 10 MHz, 20 MHz, 40 MHz, 80 MHz (with contiguous bandwidths) or 80+80 MHz (160 MHz) (with non-contiguous bandwidths). In some embodiments, a 920 MHz channel bandwidth may be used. The scope of the embodiments is not limited with respect to the above center frequencies however.

FIG. 10 illustrates WLAN FEM circuitry **904a** in accordance with some embodiments. Although the example of FIG. 10 is described in conjunction with the WLAN FEM circuitry **904a**, the example of FIG. 10 may be described in conjunction with the example BT FEM circuitry **904b** (FIG. 9), although other circuitry configurations may also be suitable.

In some embodiments, the FEM circuitry **904a** may include a TX/RX switch **1002** to switch between transmit mode and receive mode operation. The FEM circuitry **904a** may include a receive signal path and a transmit signal path. The receive signal path of the FEM circuitry **904a** may include a low-noise amplifier (LNA) **1006** to amplify received RF signals **1003** and provide the amplified received RF signals **1007** as an output (e.g., to the radio IC circuitry **906a-b** (FIG. 9)). The transmit signal path of the circuitry **904a** may include a power amplifier (PA) to amplify input RF signals **1009** (e.g., provided by the radio IC circuitry **906a-b**), and one or more filters **1012**, such as band-pass filters (BPFs), low-pass filters (LPFs) or other types of filters, to generate RF signals **1015** for subsequent transmission (e.g., by one or more of the antennas **901** (FIG. 9)) via an example duplexer **1014**.

In some dual-mode embodiments for Wi-Fi communication, the FEM circuitry **904a** may be configured to operate in either the 2.4 GHz frequency spectrum or the 5 GHz frequency spectrum. In these embodiments, the receive signal path of the FEM circuitry **904a** may include a receive signal path duplexer **1004** to separate the signals from each spectrum as well as provide a separate LNA **1006** for each spectrum as shown. In these embodiments, the transmit signal path of the FEM circuitry **904a** may also include a power amplifier **1010** and a filter **1012**, such as a BPF, an LPF or another type of filter for each frequency spectrum and a transmit signal path duplexer **1004** to provide the signals of one of the different spectrums onto a single transmit path for subsequent transmission by the one or more of the antennas **901** (FIG. 9). In some embodiments, BT communications may utilize the 2.4 GHz signal paths and may utilize the same FEM circuitry **904a** as the one used for WLAN communications.

FIG. 11 illustrates radio IC circuitry **906a** in accordance with some embodiments. The radio IC circuitry **906a** is one example of circuitry that may be suitable for use as the WLAN or BT radio IC circuitry **906a/906b** (FIG. 9), although other circuitry configurations may also be suitable. Alternatively, the example of FIG. 11 may be described in conjunction with the example BT radio IC circuitry **906b**.

In some embodiments, the radio IC circuitry **906a** may include a receive signal path and a transmit signal path. The receive signal path of the radio IC circuitry **906a** may include at least mixer circuitry **1102**, such as, for example, down-conversion mixer circuitry, amplifier circuitry **1106** and filter circuitry **1108**. The transmit signal path of the radio IC circuitry **906a** may include at least filter circuitry **1112** and mixer circuitry **1114**, such as, for example, up-conversion mixer circuitry. Radio IC circuitry **906a** may also include synthesizer circuitry **1104** for synthesizing a frequency **1105** for use by the mixer circuitry **1102** and the mixer circuitry **1114**. The mixer circuitry **1102** and/or **1114** may each, according to some embodiments, be configured to provide direct conversion functionality. The latter type of circuitry presents a much simpler architecture as compared with standard super-heterodyne mixer circuitries, and any flicker noise brought about by the same may be alleviated for example through the use of OFDM modulation. FIG. 11 illustrates only a simplified version of a radio IC circuitry, and may include, although not shown, embodiments where each of the depicted circuitries may include more than one component. For instance, mixer circuitry **1114** may each include one or more mixers, and filter circuitries **1108** and/or **1112** may each include one or more filters, such as one or more BPFs and/or LPFs according to application needs. For example, when mixer circuitries are of the direct-conversion type, they may each include two or more mixers.

In some embodiments, mixer circuitry **1102** may be configured to down-convert RF signals **1007** received from the FEM circuitry **904a-b** (FIG. 9) based on the synthesized frequency **1105** provided by synthesizer circuitry **1104**. The amplifier circuitry **1106** may be configured to amplify the down-converted signals and the filter circuitry **1108** may include an LPF configured to remove unwanted signals from the down-converted signals to generate output baseband signals **1107**. Output baseband signals **1107** may be provided to the baseband processing circuitry **908a-b** (FIG. 9) for further processing. In some embodiments, the output baseband signals **1107** may be zero-frequency baseband signals, although this is not a requirement. In some embodi-

ments, mixer circuitry **1102** may comprise passive mixers, although the scope of the embodiments is not limited in this respect.

In some embodiments, the mixer circuitry **1114** may be configured to up-convert input baseband signals **1111** based on the synthesized frequency **1105** provided by the synthesizer circuitry **1104** to generate RF output signals **1009** for the FEM circuitry **904a-b**. The baseband signals **1111** may be provided by the baseband processing circuitry **908a-b** and may be filtered by filter circuitry **1112**. The filter circuitry **1112** may include an LPF or a BPF, although the scope of the embodiments is not limited in this respect.

In some embodiments, the mixer circuitry **1102** and the mixer circuitry **1114** may each include two or more mixers and may be arranged for quadrature down-conversion and/or up-conversion respectively with the help of synthesizer **1104**. In some embodiments, the mixer circuitry **1102** and the mixer circuitry **1114** may each include two or more mixers each configured for image rejection (e.g., Hartley image rejection). In some embodiments, the mixer circuitry **1102** and the mixer circuitry **1114** may be arranged for direct down-conversion and/or direct up-conversion, respectively. In some embodiments, the mixer circuitry **1102** and the mixer circuitry **1114** may be configured for super-heterodyne operation, although this is not a requirement.

Mixer circuitry **1102** may comprise, according to one embodiment: quadrature passive mixers (e.g., for the in-phase (I) and quadrature phase (Q) paths). In such an embodiment, RF input signal **1007** from FIG. 11 may be down-converted to provide I and Q baseband output signals to be sent to the baseband processor.

Quadrature passive mixers may be driven by zero and ninety-degree time-varying LO switching signals provided by a quadrature circuitry which may be configured to receive a LO frequency (f_{LO}) from a local oscillator or a synthesizer, such as LO frequency **1105** of synthesizer **1104** (FIG. 11). In some embodiments, the LO frequency may be the carrier frequency, while in other embodiments, the LO frequency may be a fraction of the carrier frequency (e.g., one-half the carrier frequency, one-third the carrier frequency). In some embodiments, the zero and ninety-degree time-varying switching signals may be generated by the synthesizer, although the scope of the embodiments is not limited in this respect.

In some embodiments, the LO signals may differ in duty cycle (the percentage of one period in which the LO signal is high) and/or offset (the difference between start points of the period). In some embodiments, the LO signals may have an 85% duty cycle and an 80% offset. In some embodiments, each branch of the mixer circuitry (e.g., the in-phase (I) and quadrature phase (Q) path) may operate at an 80% duty cycle, which may result in a significant reduction in power consumption.

The RF input signal **1007** (FIG. 10) may comprise a balanced signal, although the scope of the embodiments is not limited in this respect. The I and Q baseband output signals may be provided to low-noise amplifier, such as amplifier circuitry **1106** (FIG. 11) or to filter circuitry **1108** (FIG. 11).

In some embodiments, the output baseband signals **1107** and the input baseband signals **1111** may be analog baseband signals, although the scope of the embodiments is not limited in this respect. In some alternate embodiments, the output baseband signals **1107** and the input baseband signals **1111** may be digital baseband signals. In these alternate

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embodiments, the radio IC circuitry may include analog-to-digital converter (ADC) and digital-to-analog converter (DAC) circuitry.

In some dual-mode embodiments, a separate radio IC circuitry may be provided for processing signals for each spectrum, or for other spectrums not mentioned here, although the scope of the embodiments is not limited in this respect.

In some embodiments, the synthesizer circuitry **1104** may be a fractional-N synthesizer or a fractional N/N+1 synthesizer, although the scope of the embodiments is not limited in this respect as other types of frequency synthesizers may be suitable. For example, synthesizer circuitry **1104** may be a delta-sigma synthesizer, a frequency multiplier, or a synthesizer comprising a phase-locked loop with a frequency divider. According to some embodiments, the synthesizer circuitry **1104** may include digital synthesizer circuitry. An advantage of using a digital synthesizer circuitry is that, although it may still include some analog components, its footprint may be scaled down much more than the footprint of an analog synthesizer circuitry. In some embodiments, frequency input into synthesizer circuitry **1104** may be provided by a voltage controlled oscillator (VCO), although that is not a requirement. A divider control input may further be provided by either the baseband processing circuitry **908a-b** (FIG. 9) depending on the desired output frequency **1105**. In some embodiments, a divider control input (e.g., N) may be determined from a look-up table (e.g., within a Wi-Fi card) based on a channel number and a channel center frequency as determined or indicated by the example application processor **910**. The application processor **910** may include, or otherwise be connected to, one of the example secure signal converter **101** or the example received signal converter **103** (e.g., depending on which device the example radio architecture is implemented in).

In some embodiments, synthesizer circuitry **1104** may be configured to generate a carrier frequency as the output frequency **1105**, while in other embodiments, the output frequency **1105** may be a fraction of the carrier frequency (e.g., one-half the carrier frequency, one-third the carrier frequency). In some embodiments, the output frequency **1105** may be a LO frequency (fLO).

FIG. 12 illustrates a functional block diagram of baseband processing circuitry **908a** in accordance with some embodiments. The baseband processing circuitry **908a** is one example of circuitry that may be suitable for use as the baseband processing circuitry **908a** (FIG. 9), although other circuitry configurations may also be suitable. Alternatively, the example of FIG. 11 may be used to implement the example BT baseband processing circuitry **908b** of FIG. 9.

The baseband processing circuitry **908a** may include a receive baseband processor (RX BBP) **1202** for processing receive baseband signals **1109** provided by the radio IC circuitry **906a-b** (FIG. 9) and a transmit baseband processor (TX BBP) **1204** for generating transmit baseband signals **1111** for the radio IC circuitry **906a-b**. The baseband processing circuitry **908a** may also include control logic **1206** for coordinating the operations of the baseband processing circuitry **908a**.

In some embodiments (e.g., when analog baseband signals are exchanged between the baseband processing circuitry **908a-b** and the radio IC circuitry **906a-b**), the baseband processing circuitry **908a** may include ADC **1210** to convert analog baseband signals **1209** received from the radio IC circuitry **906a-b** to digital baseband signals for processing by the RX BBP **1202**. In these embodiments, the baseband processing circuitry **908a** may also include DAC

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1212 to convert digital baseband signals from the TX BBP **1204** to analog baseband signals **1211**.

In some embodiments that communicate OFDM signals or OFDMA signals, such as through baseband processor **908a**, the transmit baseband processor **1204** may be configured to generate OFDM or OFDMA signals as appropriate for transmission by performing an inverse fast Fourier transform (IFFT). The receive baseband processor **1202** may be configured to process received OFDM signals or OFDMA signals by performing an FFT. In some embodiments, the receive baseband processor **1202** may be configured to detect the presence of an OFDM signal or OFDMA signal by performing an autocorrelation, to detect a preamble, such as a short preamble, and by performing a cross-correlation, to detect a long preamble. The preambles may be part of a predetermined frame structure for Wi-Fi communication.

Referring back to FIG. 9, in some embodiments, the antennas **901** (FIG. 9) may each comprise one or more directional or omnidirectional antennas, including, for example, dipole antennas, monopole antennas, patch antennas, loop antennas, microstrip antennas or other types of antennas suitable for transmission of RF signals. In some multiple-input multiple-output (MIMO) embodiments, the antennas may be effectively separated to take advantage of spatial diversity and the different channel characteristics that may result. Antennas **901** may each include a set of phased-array antennas, although embodiments are not so limited.

Although the radio architecture **105A**, **105B** is illustrated as having several separate functional elements, one or more of the functional elements may be combined and may be implemented by combinations of software-configured elements, such as processing elements including digital signal processors (DSPs), and/or other hardware elements. For example, some elements may comprise one or more microprocessors, DSPs, field-programmable gate arrays (FPGAs), application specific integrated circuits (ASICs), radio-frequency integrated circuits (RFICs) and combinations of various hardware and logic circuitry for performing at least the functions described herein. In some embodiments, the functional elements may refer to one or more processes operating on one or more processing elements.

The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments. The terms “computing device,” “user device,” “communication station,” “station,” “handheld device,” “mobile device,” “wireless device” and “user equipment” (UE) as used herein refers to a wireless communication device such as a cellular telephone, a smartphone, a tablet, a netbook, a wireless terminal, a laptop computer, a femtocell, a high data rate (HDR) subscriber station, an access point, a printer, a point of sale device, an access terminal, or other personal communication system (PCS) device. The device may be either mobile or stationary.

As used within this document, the term “communicate” is intended to include transmitting, or receiving, or both transmitting and receiving. This may be particularly useful in claims when describing the organization of data that is being transmitted by one device and received by another, but only the functionality of one of those devices is required to infringe the claim. Similarly, the bidirectional exchange of data between two devices (both devices transmit and receive during the exchange) may be described as “communicating,” when only the functionality of one of those devices is being claimed. The term “communicating” as used herein

with respect to a wireless communication signal includes transmitting the wireless communication signal and/or receiving the wireless communication signal. For example, a wireless communication unit, which is capable of communicating a wireless communication signal, may include a wireless transmitter to transmit the wireless communication signal to at least one other wireless communication unit, and/or a wireless communication receiver to receive the wireless communication signal from at least one other wireless communication unit.

As used herein, unless otherwise specified, the use of the ordinal adjectives “first,” “second,” “third,” etc., to describe a common object, merely indicates that different instances of like objects are being referred to and are not intended to imply that the objects so described must be in a given sequence, either temporally, spatially, in ranking, or in any other manner.

The term “access point” (AP) as used herein may be a fixed station. An access point may also be referred to as an access node, a base station, an evolved node B (eNodeB), or some other similar terminology known in the art. An access terminal may also be called a mobile station, user equipment (UE), a wireless communication device, or some other similar terminology known in the art. Embodiments disclosed herein generally pertain to wireless networks. Some embodiments may relate to wireless networks that operate in accordance with one of the IEEE 802.11 standards.

Some embodiments may be used in conjunction with various devices and systems, for example, a personal computer (PC), a desktop computer, a mobile computer, a laptop computer, a notebook computer, a tablet computer, a server computer, a handheld computer, a handheld device, a personal digital assistant (PDA) device, a handheld PDA device, an on-board device, an off-board device, a hybrid device, a vehicular device, a non-vehicular device, a mobile or portable device, a consumer device, a non-mobile or non-portable device, a wireless communication station, a wireless communication device, a wireless access point (AP), a wired or wireless router, a wired or wireless modem, a video device, an audio device, an audio-video (A/V) device, a wired or wireless network, a wireless area network, a wireless video area network (WVAN), a local area network (LAN), a wireless LAN (WLAN), a personal area network (PAN), a wireless PAN (WPAN), and the like.

Some embodiments may be used in conjunction with one way and/or two-way radio communication systems, cellular radio-telephone communication systems, a mobile phone, a cellular telephone, a wireless telephone, a personal communication system (PCS) device, a PDA device which incorporates a wireless communication device, a mobile or portable global positioning system (GPS) device, a device which incorporates a GPS receiver or transceiver or chip, a device which incorporates an RFID element or chip, a multiple input multiple output (MIMO) transceiver or device, a single input multiple output (SIMO) transceiver or device, a multiple input single output (MISO) transceiver or device, a device having one or more internal antennas and/or external antennas, digital video broadcast (DVB) devices or systems, multi-standard radio devices or systems, a wired or wireless handheld device, e.g., a smartphone, a wireless application protocol (WAP) device, or the like.

Some embodiments may be used in conjunction with one or more types of wireless communication signals and/or systems following one or more wireless communication protocols, for example, radio frequency (RF), infrared (IR), frequency-division multiplexing (FDM), orthogonal FDM (OFDM), time-division multiplexing (TDM), time-division

multiple access (TDMA), extended TDMA (E-TDMA), general packet radio service (GPRS), extended GPRS, code-division multiple access (CDMA), wideband CDMA (WCDMA), CDMA 2000, single-carrier CDMA, multi-carrier CDMA, multi-carrier modulation (MDM), discrete multi-tone (DMT), Bluetooth®, global positioning system (GPS), Wi-Fi, Wi-Max, ZigBee, ultra-wideband (UWB), global system for mobile communications (GSM), 2G, 2.5G, 3G, 3.5G, 4G, fifth generation (5G) mobile networks, 3GPP, long term evolution (LTE), LTE advanced, enhanced data rates for GSM Evolution (EDGE), or the like. Other embodiments may be used in various other devices, systems, and/or networks.

The following examples pertain to further embodiments.

Example 1 may include a device comprising processing circuitry coupled to storage, the processing circuitry configured to: utilize a legacy baseband data for a generation of a plurality of orthogonal frequency-division multiplexing (OFDM) symbols for a 60 GHz communication; select a number of subcarriers based on a bandwidth selection; generate an upclock of the legacy baseband data by an upclocking factor that aligns a first bandwidth with a legacy bandwidth; assign a first subcarrier spacing for a first group of bandwidths; assign a second subcarrier spacing for a second group of bandwidths; generate the plurality of OFDM symbols using an upclocking mechanism based on a bandwidth for the 60 GHz communication; and cause to send a plurality of OFDM symbols to a station device.

Example 2 may include the device of example 1 and/or some other example herein, wherein the number of subcarriers may be at least one of 64, 128, 256, or 512.

Example 3 may include the device of example 1 and/or some other example herein, wherein the legacy baseband data may be associated with a 20 MHz frequency for IEEE 802.11ac.

Example 4 may include the device of example 1 and/or some other example herein, wherein a legacy subcarriers spacing may be 312.5 kHz.

Example 5 may include the device of example 1 and/or some other example herein, wherein the upclocking factor may be 6.75.

Example 6 may include the device of example 1 and/or some other example herein, wherein the processing circuitry may be further configured to upclock the legacy baseband data by 6.75 times for a 135 MHz bandwidth in the 60 GHz communication.

Example 7 may include the device of example 1 and/or some other example herein, wherein the processing circuitry may be further configured to: utilize a first legacy data for a 40 MHz physical layer (PHY) convergence protocol data unit (PPDU); upclock the first legacy data by 6.75 times for a 270 MHz bandwidth for the 60 GHz communication.

Example 8 may include the device of example 1 and/or some other example herein, wherein the processing circuitry may be further configured to: utilize a first legacy data for a 80 MHz physical layer (PHY) convergence protocol data unit (PPDU); upclock the first legacy data by 6.75 times for a 540 MHz bandwidth for the 60 GHz communication.

Example 9 may include the device of example 1 and/or some other example herein, wherein the processing circuitry may be further configured to: utilize a first legacy data for a 160 MHz physical layer (PHY) convergence protocol data unit (PPDU); upclock the first legacy data by 6.75 times for a 1080 MHz bandwidth for the 60 GHz communication.

Example 10 may include a non-transitory computer-readable medium storing computer-executable instructions which when executed by one or more processors result in

performing operations comprising: utilizing a legacy baseband data for a generation of a plurality of orthogonal frequency-division multiplexing (OFDM) symbols for a 60 GHz communication; selecting a number of subcarriers based on a bandwidth selection; generating an upclock of the legacy baseband data by an upclocking factor that aligns a first bandwidth with a legacy bandwidth; assigning a first subcarrier spacing for a first group of bandwidths; assigning a second subcarrier spacing for a second group of bandwidths; generating the plurality of OFDM symbols using an upclocking mechanism based on a bandwidth for the 60 GHz communication; and causing to send a plurality of OFDM symbols to a station device.

Example 11 may include the non-transitory computer-readable medium of example 10 and/or some other example herein, wherein the number of subcarriers may be at least one of 64, 128, 256, or 512.

Example 12 may include the non-transitory computer-readable medium of example 10 and/or some other example herein, wherein the legacy baseband data may be associated with a 20 MHz frequency for IEEE 802.11ac.

Example 13 may include the non-transitory computer-readable medium of example 10 and/or some other example herein, wherein a legacy subcarriers spacing may be 312.5 kHz.

Example 14 may include the non-transitory computer-readable medium of example 10 and/or some other example herein, wherein the upclocking factor may be 6.75.

Example 15 may include the non-transitory computer-readable medium of example 10 and/or some other example herein, wherein the operations further comprise upclock the legacy baseband data by 6.75 times for a 135 MHz bandwidth in the 60 GHz communication.

Example 16 may include the non-transitory computer-readable medium of example 10 and/or some other example herein, wherein the operations further comprise: utilizing a first legacy data for a 40 MHz physical layer (PHY) convergence protocol data unit (PPDU); upclock the first legacy data by 6.75 times for a 270 MHz bandwidth for the 60 GHz communication.

Example 17 may include the non-transitory computer-readable medium of example 10 and/or some other example herein, wherein the operations further comprise: utilizing a first legacy data for a 80 MHz physical layer (PHY) convergence protocol data unit (PPDU); upclock the first legacy data by 6.75 times for a 540 MHz bandwidth for the 60 GHz communication.

Example 18 may include the non-transitory computer-readable medium of example 10 and/or some other example herein, wherein the operations further comprise: utilizing a first legacy data for a 160 MHz physical layer (PHY) convergence protocol data unit (PPDU); upclock the first legacy data by 6.75 times for a 1080 MHz bandwidth for the 60 GHz communication.

Example 19 may include a method comprising: utilizing a legacy baseband data for a generation of a plurality of orthogonal frequency-division multiplexing (OFDM) symbols for a 60 GHz communication; selecting a number of subcarriers based on a bandwidth selection; generating an upclock of the legacy baseband data by an upclocking factor that aligns a first bandwidth with a legacy bandwidth; assigning a first subcarrier spacing for a first group of bandwidths; assigning a second subcarrier spacing for a second group of bandwidths; generating the plurality of OFDM symbols using an upclocking mechanism based on a bandwidth for the 60 GHz communication; and causing to send a plurality of OFDM symbols to a station device.

Example 20 may include the method of example 18 and/or some other example herein, wherein the number of subcarriers may be at least one of 64, 128, 256, or 512.

Example 21 may include the method of example 18 and/or some other example herein, wherein the legacy baseband data may be associated with a 20 MHz frequency for IEEE 802.11ac.

Example 22 may include the method of example 18 and/or some other example herein, wherein a legacy subcarriers spacing may be 312.5 kHz.

Example 23 may include the method of example 18 and/or some other example herein, wherein the upclocking factor may be 6.75.

Example 24 may include the method of example 18 and/or some other example herein, further comprising upclock the legacy baseband data by 6.75 times for a 135 MHz bandwidth in the 60 GHz communication.

Example 25 may include the method of example 18 and/or some other example herein, further comprising: utilizing a first legacy data for a 40 MHz physical layer (PHY) convergence protocol data unit (PPDU); upclock the first legacy data by 6.75 times for a 270 MHz bandwidth for the 60 GHz communication.

Example 26 may include the method of example 18 and/or some other example herein, further comprising: utilizing a first legacy data for a 80 MHz physical layer (PHY) convergence protocol data unit (PPDU); upclock the first legacy data by 6.75 times for a 540 MHz bandwidth for the 60 GHz communication.

Example 27 may include the method of example 18 and/or some other example herein, further comprising: utilizing a first legacy data for a 160 MHz physical layer (PHY) convergence protocol data unit (PPDU); upclock the first legacy data by 6.75 times for a 1080 MHz bandwidth for the 60 GHz communication.

Example 28 may include an apparatus comprising means for: utilizing a legacy baseband data for a generation of a plurality of orthogonal frequency-division multiplexing (OFDM) symbols for a 60 GHz communication; selecting a number of subcarriers based on a bandwidth selection; generating an upclock of the legacy baseband data by an upclocking factor that aligns a first bandwidth with a legacy bandwidth; assigning a first subcarrier spacing for a first group of bandwidths; assigning a second subcarrier spacing for a second group of bandwidths; generating the plurality of OFDM symbols using an upclocking mechanism based on a bandwidth for the 60 GHz communication; and causing to send a plurality of OFDM symbols to a station device.

Example 29 may include the apparatus of example 27 and/or some other example herein, wherein the number of subcarriers may be at least one of 64, 128, 256, or 512.

Example 30 may include the apparatus of example 27 and/or some other example herein, wherein the legacy baseband data may be associated with a 20 MHz frequency for IEEE 802.11ac.

Example 31 may include the apparatus of example 27 and/or some other example herein, wherein a legacy subcarriers spacing may be 312.5 kHz.

Example 32 may include the apparatus of example 27 and/or some other example herein, wherein the upclocking factor may be 6.75.

Example 33 may include the apparatus of example 27 and/or some other example herein, further comprising upclock the legacy baseband data by 6.75 times for a 135 MHz bandwidth in the 60 GHz communication.

Example 34 may include the apparatus of example 27 and/or some other example herein, further comprising: uti-

lizing a first legacy data for a 40 MHz physical layer (PHY) convergence protocol data unit (PPDU); upclock the first legacy data by 6.75 times for a 270 MHz bandwidth for the 60 GHz communication.

Example 35 may include the apparatus of example 27 and/or some other example herein, further comprising: utilizing a first legacy data for a 80 MHz physical layer (PHY) convergence protocol data unit (PPDU); upclock the first legacy data by 6.75 times for a 540 MHz bandwidth for the 60 GHz communication.

Example 36 may include the apparatus of example 27 and/or some other example herein, further comprising: utilizing a first legacy data for a 160 MHz physical layer (PHY) convergence protocol data unit (PPDU); upclock the first legacy data by 6.75 times for a 1080 MHz bandwidth for the 60 GHz communication.

Example 37 may include one or more non-transitory computer-readable media comprising instructions to cause an electronic device, upon execution of the instructions by one or more processors of the electronic device, to perform one or more elements of a method described in or related to any of examples 1-36, or any other method or process described herein.

Example 38 may include an apparatus comprising logic, modules, and/or circuitry to perform one or more elements of a method described in or related to any of examples 1-36, or any other method or process described herein.

Example 39 may include a method, technique, or process as described in or related to any of examples 1-36, or portions or parts thereof.

Example 40 may include an apparatus comprising: one or more processors and one or more computer readable media comprising instructions that, when executed by the one or more processors, cause the one or more processors to perform the method, techniques, or process as described in or related to any of examples 1-36, or portions thereof.

Example 40 may include a method of communicating in a wireless network as shown and described herein.

Example 42 may include a system for providing wireless communication as shown and described herein.

Example 43 may include a device for providing wireless communication as shown and described herein.

Embodiments according to the disclosure are in particular disclosed in the attached claims directed to a method, a storage medium, a device and a computer program product, wherein any feature mentioned in one claim category, e.g., method, can be claimed in another claim category, e.g., system, as well. The dependencies or references back in the attached claims are chosen for formal reasons only. However, any subject matter resulting from a deliberate reference back to any previous claims (in particular multiple dependencies) can be claimed as well, so that any combination of claims and the features thereof are disclosed and can be claimed regardless of the dependencies chosen in the attached claims. The subject-matter which can be claimed comprises not only the combinations of features as set out in the attached claims but also any other combination of features in the claims, wherein each feature mentioned in the claims can be combined with any other feature or combination of other features in the claims. Furthermore, any of the embodiments and features described or depicted herein can be claimed in a separate claim and/or in any combination with any embodiment or feature described or depicted herein or with any of the features of the attached claims.

The foregoing description of one or more implementations provides illustration and description, but is not intended to be exhaustive or to limit the scope of embodi-

ments to the precise form disclosed. Modifications and variations are possible in light of the above teachings or may be acquired from practice of various embodiments.

Certain aspects of the disclosure are described above with reference to block and flow diagrams of systems, methods, apparatuses, and/or computer program products according to various implementations. It will be understood that one or more blocks of the block diagrams and flow diagrams, and combinations of blocks in the block diagrams and the flow diagrams, respectively, may be implemented by computer-executable program instructions. Likewise, some blocks of the block diagrams and flow diagrams may not necessarily need to be performed in the order presented, or may not necessarily need to be performed at all, according to some implementations.

These computer-executable program instructions may be loaded onto a special-purpose computer or other particular machine, a processor, or other programmable data processing apparatus to produce a particular machine, such that the instructions that execute on the computer, processor, or other programmable data processing apparatus create means for implementing one or more functions specified in the flow diagram block or blocks. These computer program instructions may also be stored in a computer-readable storage media or memory that may direct a computer or other programmable data processing apparatus to function in a particular manner, such that the instructions stored in the computer-readable storage media produce an article of manufacture including instruction means that implement one or more functions specified in the flow diagram block or blocks. As an example, certain implementations may provide for a computer program product, comprising a computer-readable storage medium having a computer-readable program code or program instructions implemented therein, said computer-readable program code adapted to be executed to implement one or more functions specified in the flow diagram block or blocks. The computer program instructions may also be loaded onto a computer or other programmable data processing apparatus to cause a series of operational elements or steps to be performed on the computer or other programmable apparatus to produce a computer-implemented process such that the instructions that execute on the computer or other programmable apparatus provide elements or steps for implementing the functions specified in the flow diagram block or blocks.

Accordingly, blocks of the block diagrams and flow diagrams support combinations of means for performing the specified functions, combinations of elements or steps for performing the specified functions and program instruction means for performing the specified functions. It will also be understood that each block of the block diagrams and flow diagrams, and combinations of blocks in the block diagrams and flow diagrams, may be implemented by special-purpose, hardware-based computer systems that perform the specified functions, elements or steps, or combinations of special-purpose hardware and computer instructions.

Conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain implementations could include, while other implementations do not include, certain features, elements, and/or operations. Thus, such conditional language is not generally intended to imply that features, elements, and/or operations are in any way required for one or more implementations or that one or more implementations necessarily include logic for deciding, with or without user input or prompting, whether these features,

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elements, and/or operations are included or are to be performed in any particular implementation.

Many modifications and other implementations of the disclosure set forth herein will be apparent having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the disclosure is not to be limited to the specific implementations disclosed and that modifications and other implementations are intended to be included within the scope of the appended claims. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

What is claimed is:

1. A device, the device comprising processing circuitry coupled to storage, the processing circuitry configured to: utilize a legacy baseband data for a generation of a plurality of orthogonal frequency-division multiplexing (OFDM) symbols for a 60 GHz communication; select a number of subcarriers based on a bandwidth selection;

generate an upclock of the legacy baseband data by an upclocking factor that aligns a first bandwidth with a legacy bandwidth;

assign a first subcarrier spacing for a first group of bandwidths;

assign a second subcarrier spacing for a second group of bandwidths;

generate the plurality of OFDM symbols using an upclocking mechanism based on a bandwidth for the 60 GHz communication; and

cause to send a plurality of OFDM symbols to a station device.

2. The device of claim 1, wherein the number of subcarriers is at least one of 64, 128, 256, or 512.

3. The device of claim 1, wherein the legacy baseband data is associated with a 20 MHz frequency for IEEE 802.11ac.

4. The device of claim 1, wherein a legacy subcarriers spacing is 312.5 kHz.

5. The device of claim 1, wherein the upclocking factor is 6.75.

6. The device of claim 1, wherein the processing circuitry is further configured to upclock the legacy baseband data by 6.75 times for a 135 MHz bandwidth in the 60 GHz communication.

7. The device of claim 1, wherein the processing circuitry is further configured to:

utilize a first legacy data for a 40 MHz physical layer (PHY) convergence protocol data unit (PPDU);

upclock the first legacy data by 6.75 times for a 270 MHz bandwidth for the 60 GHz communication.

8. The device of claim 1, wherein the processing circuitry is further configured to:

utilize a first legacy data for a 80 MHz physical layer (PHY) convergence protocol data unit (PPDU);

upclock the first legacy data by 6.75 times for a 540 MHz bandwidth for the 60 GHz communication.

9. The device of claim 1, wherein the processing circuitry is further configured to:

utilize a first legacy data for a 160 MHz physical layer (PHY) convergence protocol data unit (PPDU);

upclock the first legacy data by 6.75 times for a 1080 MHz bandwidth for the 60 GHz communication.

10. A non-transitory computer-readable medium storing computer-executable instructions which when executed by one or more processors result in performing operations comprising:

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utilizing a legacy baseband data for a generation of a plurality of orthogonal frequency-division multiplexing (OFDM) symbols for a 60 GHz communication; selecting a number of subcarriers based on a bandwidth selection;

generating an upclock of the legacy baseband data by an upclocking factor that aligns a first bandwidth with a legacy bandwidth;

assigning a first subcarrier spacing for a first group of bandwidths;

assigning a second subcarrier spacing for a second group of bandwidths;

generating the plurality of OFDM symbols using an upclocking mechanism based on a bandwidth for the 60 GHz communication; and

causing to send a plurality of OFDM symbols to a station device.

11. The non-transitory computer-readable medium of claim 10, wherein the number of subcarriers is at least one of 64, 128, 256, or 512.

12. The non-transitory computer-readable medium of claim 10, wherein the legacy baseband data is associated with a 20 MHz frequency for IEEE 802.11ac.

13. The non-transitory computer-readable medium of claim 10, wherein a legacy subcarriers spacing is 312.5 kHz.

14. The non-transitory computer-readable medium of claim 10, wherein the upclocking factor is 6.75.

15. The non-transitory computer-readable medium of claim 10, wherein the operations further comprise upclock the legacy baseband data by 6.75 times for a 135 MHz bandwidth in the 60 GHz communication.

16. The non-transitory computer-readable medium of claim 10, wherein the operations further comprise:

utilizing a first legacy data for a 40 MHz physical layer (PHY) convergence protocol data unit (PPDU);

upclock the first legacy data by 6.75 times for a 270 MHz bandwidth for the 60 GHz communication.

17. The non-transitory computer-readable medium of claim 10, wherein the operations further comprise:

utilizing a first legacy data for a 80 MHz physical layer (PHY) convergence protocol data unit (PPDU);

upclock the first legacy data by 6.75 times for a 540 MHz bandwidth for the 60 GHz communication.

18. The non-transitory computer-readable medium of claim 10, wherein the operations further comprise:

utilizing a first legacy data for a 160 MHz physical layer (PHY) convergence protocol data unit (PPDU);

upclock the first legacy data by 6.75 times for a 1080 MHz bandwidth for the 60 GHz communication.

19. The method of claim 18, wherein the number of subcarriers is at least one of 64, 128, 256, or 512.

20. A method comprising:

utilizing a legacy baseband data for a generation of a plurality of orthogonal frequency-division multiplexing (OFDM) symbols for a 60 GHz communication; selecting a number of subcarriers based on a bandwidth selection;

generating an upclock of the legacy baseband data by an upclocking factor that aligns a first bandwidth with a legacy bandwidth;

assigning a first subcarrier spacing for a first group of bandwidths;

assigning a second subcarrier spacing for a second group of bandwidths;

generating the plurality of OFDM symbols using an upclocking mechanism based on a bandwidth for the 60 GHz communication; and

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causing to send a plurality of OFDM symbols to a station
device.

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